

CHRONUS: A MEMORY-TRAINED AGENTIC AI FOR HUMAN PERSONALITY AND MINDSET SIMULATION

A Project Report

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CERTIFICATE

This is to certify that Project – II (303105423) of 7th Semester entitled “**CHRONUS: A Memory-Trained Agentic AI for Human Personality and Mindset Simulation**” of Group No. CSE_235 has been successfully completed by

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“The single greatest cause of happiness is gratitude.”

–Auliq-Ice

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Abstract

Human personality, memory, and communication style are among the most fragile aspects of individual identity. Once a person passes away or becomes cognitively impaired, their unique worldview, values, and voice are irreversibly lost. Existing digital legacy platforms either store static archives without interactive capability or rely on unconstrained large language models that frequently hallucinate autobiographical facts, fabricate memories, and produce persona-inconsistent responses.

CHRONUS addresses this challenge by proposing a memory-trained agentic AI system designed for evidence-based simulation of documented human personality and mindset. The system ingests personal documents—including text files, letters, PDFs, emails, and structured interviews—and segments them into atomic memory units using spaCy-based natural language processing. Each unit is encoded into a dense 384-dimensional vector representation using Sentence-BERT (all-MiniLM-L6-v2) and stored in a locally persistent ChromaDB vector database employing HNSW-based approximate nearest-neighbour indexing.

At query time, CHRONUS performs cosine-similarity retrieval of the top-3 most semantically relevant memory units and constructs responses using the Mix Method: a three-part generation strategy comprising a persona-consistent framing, an evidentiary core grounded in retrieved authentic passages, and a controlled closing signature. A structured 25-question interview augmentation protocol captures six personality dimensions—personality traits, core memories, relationships, passions, beliefs and values, and voice and communication style—to overcome the natural sparsity of unstructured personal archives.

The system supports both cloud-based voice synthesis via ElevenLabs and privacy-preserving local voice cloning through XTTS-v2. The architecture follows a local-first, privacy-first design principle, with all embedding computation, vector storage, and retrieval executed on-device without transmitting personal data to external servers.

Empirical evaluation demonstrates that CHRONUS achieves a Precision@1 of 0.82 for semantic retrieval compared to 0.41 for keyword-based baselines, a groundedness score of 4.6/5.0 versus 2.3/5.0 for unconstrained generative approaches, and a safety score of 4.8/5.0 versus 3.5/5.0. The 25-question interview augmentation increases persona coverage from 0.78 to 0.94 and emotional relevance from 2.9 to 4.0 on a five-point scale.

The system includes eight pre-trained famous-personality models for demonstration purposes and provides a web-based interface built with React 18, TypeScript, and Vite, backed by a Python 3.11 Flask API and a Node.js Express proxy layer. Current limitations and future directions—including real-time voice streaming, facial avatar generation, LLM-integrated retrieval-augmented generation, and clinically informed grief-safety protocols—are discussed.

Keywords: Personality Simulation, Memory-Grounded AI, Sentence-BERT, ChromaDB, Vector Retrieval, Agentic AI, Digital Legacy, Voice Cloning, Anti-Hallucination, Privacy-First Architecture

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Chapter 1

Introduction

1.1 Introduction

The rapid advancement of artificial intelligence and natural language processing has opened unprecedented avenues for human-computer interaction. While contemporary large language models demonstrate remarkable proficiency in general-purpose text generation, they predominantly function as generic conversational agents devoid of persistent, grounded identities. CHRONUS is introduced as a memory-trained agentic AI designed specifically for human personality and mindset simulation. The fundamental premise of CHRONUS is to bridge the gap between static biographical archives and dynamic, interactive representations of human cognition.

Preserving human identity transcends the mere cataloging of historical facts; it necessitates the encapsulation of a person’s unique worldview, communication style, and core beliefs. As individuals produce vast amounts of personal documentation throughout their lifetimes—ranging from emails and digital messages to formal writings and journals—these fragmented data sources harbor the latent structures of their personalities. CHRONUS leverages this digital footprint to reconstruct a cohesive, interactive persona. By integrating advanced semantic retrieval architectures with memory-grounded generation techniques, the system ensures that the simulated agent strictly adheres to documented realities. This paradigm shifts the focus from generalized artificial intelligence to highly specialized, persona-specific cognitive emulation, ensuring that the essence of human identity can be preserved, interacted with, and comprehensively understood long after the original cognitive framework is no longer accessible.

1.2 Scope of the Project

The scope of the CHRONUS project encompasses the development and deployment of a full-stack, local-first architecture dedicated to high-fidelity personality simulation. The system is designed to

ingest a diverse array of personal documents—specifically supporting file formats such as TXT, MD, PDF, DOCX, CSV, EML, and JSON—from which it extracts and synthesizes the core cognitive traits of the subject. The architecture is engineered to prioritize data privacy and autonomy, utilizing a local-first operational model that minimizes reliance on external data processing for core memory retrieval.

To demonstrate the efficacy and versatility of the platform, the project includes the simulation of eight historically significant figures: Albert Einstein, Mahatma Gandhi, Nikola Tesla, Marcus Aurelius, Steve Jobs, Marie Curie, Abraham Lincoln, and William Shakespeare. These implementations serve as baseline demonstrations of the system’s ability to handle diverse rhetorical styles and complex belief systems. Furthermore, the scope extends beyond text-based interaction by incorporating advanced voice synthesis capabilities. Through the integration of ElevenLabs and XTTS-v2 technologies, CHRONUS maps the generated textual responses to synthesized vocal outputs that align with the simulated individual’s historical or documented vocal characteristics, thereby delivering a multimodal interaction experience.

1.3 Aim of the Project

The primary aim of the CHRONUS project is to develop a sophisticated, memory-grounded artificial intelligence system capable of simulating a documented human personality, mindset, and communication style with high fidelity. The project seeks to move beyond the limitations of unconstrained generative models by grounding every response in a verified corpus of personal data through semantic retrieval mechanisms. By structurally mapping a subject’s cognitive landscape—encompassing their personality, core memories, relationships, passions, beliefs, and voice—the system aims to create an interactive agent that authentically reflects the specific individual rather than a generalized archetype. Ultimately, the project endeavors to provide a robust technical framework for preserving cognitive identity, enabling future generations to engage in meaningful, contextually accurate dialogues with simulated representations of both historical figures and contemporary individuals.

1.4 Objectives of the Project

To realize the overarching aim, the CHRONUS project is guided by the following specific technical objectives:

1. **Data Ingestion Pipeline Development:** To design and implement a comprehensive data parser utilizing tools such as pdfplumber, python-docx, and pandas capable of processing

structured and unstructured formats (TXT, MD, PDF, DOCX, CSV, EML, JSON).

2. **Semantic Encoding and Storage:** To utilize Sentence-BERT (specifically the all-MiniLM-L6-v2 model with 384 dimensions) for generating dense vector representations of textual data, and to implement ChromaDB (utilizing HNSW and cosine similarity for top-3 retrieval) as the foundational vector database for efficient semantic search.
3. **Interview-Based Augmentation:** To formulate and integrate a structured 25-Question interview mechanism that systematically captures critical dimensions of the subject’s identity, including Personality, Core Memories, Relationships, Passions, Beliefs/Values, and Communication Style.
4. **Mix Method Response Generation:** To develop a hybrid prompt engineering and generation framework that synthesizes retrieved factual context with stylistic parameters to produce highly grounded, persona-consistent outputs.
5. **Voice Synthesis Integration:** To seamlessly integrate text-to-speech architectures (ElevenLabs and XTTS-v2) into the generation pipeline, allowing the simulated agent to articulate responses with appropriate vocal prosody and timbre.
6. **Privacy-First Architectural Design:** To deploy a distributed architecture—comprising a React 18, TypeScript, and Vite frontend, a Node.js/Express API proxy, and a Python 3.11 Flask backend—that ensures user data is processed securely and predominantly localized to mitigate privacy risks.

1.5 Problem Definition

The motivation for CHRONUS arises from several critical gaps in current technology regarding identity preservation and persona simulation. The problem definition is articulated through four primary domains of deficiency.

1.5.1 Irreversible Cognitive Identity Loss

Human cognition and memory are inherently ephemeral. When an individual passes away or experiences severe cognitive impairment, their unique perspective, accumulated wisdom, and specific conversational nuances are permanently lost. While their physical creations or recorded media may remain, the interactive element of their intellect is irretrievable. Society currently lacks a standardized, technological mechanism to capture the dynamic essence of a person’s cognitive

framework in a manner that allows for continued, generative interaction based strictly on their actual lived experiences and documented viewpoints.

1.5.2 The Voice Gap in AI Systems

Contemporary artificial intelligence companions often employ synthesized voices; however, these voices are typically generalized and disconnected from the specific memories and stylistic nuances of the persona being emulated. There exists a significant "voice gap" where the emotional resonance and idiosyncratic delivery of a specific individual are not faithfully reproduced. Authentic interaction requires that the synthesized voice not only matches the acoustic properties of the subject but is also intrinsically linked to the semantic content and emotional context of their personal memories.

1.5.3 Fragmented Personal Archives

In the digital age, individuals accumulate vast repositories of personal documentation, including emails, digital journals, academic papers, and social media interactions. However, these archives remain entirely static and highly fragmented. Retrieving meaningful insights or attempting to synthesize a holistic understanding of a person's worldview from these disparate sources is a manual, labor-intensive process. There is a pressing need for a system capable of autonomously aggregating, parsing, and semantically structuring this fragmented data into a unified, interactive cognitive model.

1.5.4 Hallucination in LLM-Based Persona Systems

A critical failure mode of existing Large Language Models (LLMs) when attempting persona simulation is their propensity for hallucination. Unconstrained LLMs frequently fabricate memories, invent quotes, and produce responses that are fundamentally inconsistent with the actual beliefs or historical facts of the subject. Without rigorous grounding mechanisms, these models generate plausible-sounding but entirely fictitious outputs. This lack of groundedness compromises the integrity of the simulation, making standard LLMs unsuitable for authentic identity preservation where factual accuracy and persona consistency are paramount.

1.6 Organisation of the Report

The remainder of this report is organized as follows to provide a comprehensive detailing of the CHRONUS project. Chapter 2 presents a Literature Review, examining prior work in semantic retrieval, persona-based generation, and voice synthesis. Chapter 3 discusses the System Requirements Specification, outlining the hardware and software prerequisites alongside functional

requirements. Chapter 4 details the System Design, including architectural diagrams and data flow paradigms. Chapter 5 focuses on the Implementation details, providing insights into the code structure, the data ingestion pipeline, and the Mix Method generation strategy. Chapter 6 outlines the Testing methodologies employed, including unit, integration, and system testing. Chapter 7 presents the Results and Discussion, analyzing key performance metrics such as groundedness, persona consistency, and retrieval precision. Chapter 8 highlights the Limitations of the current system. Finally, Chapter 9 concludes the report and discusses potential future enhancements. The report concludes with a comprehensive References section.

Chapter 2

Literature Survey

2.1 Introduction to the Literature Survey

The pursuit of preserving human legacy and simulating personality has historically been limited to static mediums such as written autobiographies, recorded interviews, and historical archives. However, the advent of large language models, advanced vector search algorithms, and sophisticated voice synthesis technologies has precipitated a paradigm shift towards dynamic, interactive digital preservation. This literature survey systematically reviews the foundational technologies, prior digital legacy systems, and the socio-ethical frameworks that underpin the development of CHRONUS. The review is demarcated into domains concerning existing conversational AI and legacy platforms, semantic sentence embedding, vector database architectures, neural text-to-speech synthesis, persona modelling techniques, and the critical ethical implications associated with grief technology and digital bereavement. By critically analysing both the technological capabilities and the ethical limitations of contemporary systems, this survey elucidates the research gaps that CHRONUS aims to bridge.

2.2 Conversational AI and Digital Legacy Systems

2.2.1 Replika

Replika represents a prominent example of a conversational AI companion designed primarily for emotional engagement rather than strict memory preservation. Initially conceptualised to simulate the linguistic patterns of its creator, the platform evolved into a general-purpose conversational agent [18]. While Replika demonstrates high capabilities in natural language fluency and empathetic response generation, it intrinsically lacks rigid memory grounding. Its architecture prioritises open-ended, engaging dialogue, which frequently results in hallucinatory outputs when tasked

with recalling specific factual histories. Furthermore, the absence of personalised voice cloning mechanisms restricts the multimodal fidelity of the simulation, rendering it unsuitable for high-precision digital legacy applications.

2.2.2 HereAfter AI

HereAfter AI approaches digital legacy through interactive life-story capture. The system operates on a highly structured, pre-recorded interview framework wherein users respond to specific prompts. During retrieval, the platform matches user queries to these pre-recorded video or audio segments. While this methodology entirely eliminates the risk of hallucinatory generation—as all outputs are explicit recordings of the user—it suffers from a fundamental lack of dynamic interactivity. The conversational scope is strictly confined to the pre-existing recordings, and the system cannot extrapolate or synthesise novel responses based on the user’s documented persona or beliefs.

2.2.3 StoryFile

Similar to HereAfter AI, StoryFile leverages interactive video storytelling to create conversational historical archives. It has been notably employed for Holocaust survivors and prominent historical figures, allowing users to ask questions and receive answers from pre-recorded video databases. The underlying natural language processing mechanism relies on matching the user’s question to the closest anticipated question in its repository, subsequently playing the corresponding video file. Despite its high perceptual fidelity and historical value, StoryFile is inherently pre-scripted. It does not employ semantic generative retrieval to synthesise new insights, and the absence of a generative language model prevents the system from engaging in fluid, contextually evolving dialogue outside its curated dataset.

2.2.4 Soul Machines

Soul Machines pioneers the development of autonomously animated digital people, focusing heavily on photorealistic facial animation, micro-expressions, and non-verbal communication cues. These digital entities are frequently deployed in customer service and brand representation roles. While the visual and interactive paradigms are highly advanced, the underlying conversational engines are typically generic or tailored for specific corporate knowledge bases rather than deeply grounded personal memory. The system architecture does not intrinsically support the extensive ingestion and semantic retrieval of an individual’s personal history, life events, or idiosyncratic mindset, thereby limiting its application as an authentic digital legacy medium.

2.3 Semantic Sentence Embedding and Transformer Models

2.3.1 BERT

The introduction of the Bidirectional Encoder Representations from Transformers (BERT) revolutionised natural language processing by enabling contextualised word embeddings [1]. Unlike prior models that processed text sequentially, BERT’s bidirectional architecture allows it to comprehend the context of a word based on its entire surrounding sentence. The pre-training paradigm of masked language modelling and next-sentence prediction imbued BERT with a deep understanding of linguistic structure and semantics. This foundational breakthrough established the viability of transformers for complex downstream tasks, including similarity search and information retrieval.

2.3.2 Sentence-BERT

While BERT produces robust token-level embeddings, deriving a single, meaningful fixed-size sentence embedding directly from BERT is computationally inefficient and mathematically suboptimal for similarity comparisons. Sentence-BERT (SBERT) addressed this limitation by modifying the pre-trained BERT network with siamese and triplet network structures [2]. This modification allows SBERT to derive semantically meaningful sentence embeddings that can be efficiently compared using cosine similarity. SBERT dramatically reduces the computational overhead of semantic search while maintaining high accuracy. Within the architecture of CHRONUS, SBERT (specifically the all-MiniLM-L6-v2 model) is deployed to encode both the ingested user memory corpora and the incoming user queries, facilitating rapid, highly relevant memory retrieval based on semantic meaning rather than mere keyword overlap.

2.3.3 Instruction-Tuned Models

The evolution of Large Language Models (LLMs) such as GPT-4 and LLaMA has introduced unprecedented generative capabilities. Instruction tuning has further refined these models to follow complex user directives, maintain persona constraints, and generate human-like dialogue. However, these models inherently suffer from hallucination—the generation of plausible but factually incorrect information. When unconstrained, an LLM attempting to simulate a specific individual will often invent memories or opinions to maintain conversational flow. This fundamental limitation necessitates the integration of external memory stores and strict retrieval mechanisms to ensure that the generative output remains strictly anchored to the user’s actual documented history.

2.4 Vector Databases and Similarity Search

2.4.1 Approximate Nearest Neighbour Search

The core challenge in semantic memory retrieval lies in querying vast datasets of high-dimensional vectors. Traditional exact search methods, such as k-Nearest Neighbours (k-NN), are computationally prohibitive for large-scale real-time applications due to their linear time complexity. Approximate Nearest Neighbour (ANN) algorithms mitigate this by trading a negligible degree of precision for exponential gains in search velocity. The Hierarchical Navigable Small World (HNSW) algorithm represents the state-of-the-art in ANN search, constructing a multi-layered graph that allows for logarithmic time complexity in vector retrieval.

2.4.2 ChromaDB

ChromaDB is a modern, open-source embedding database designed specifically for AI applications [7]. It provides a seamless, Python-native interface for storing, managing, and querying high-dimensional vectors and their associated metadata. CHRONUS utilises ChromaDB as its primary memory storage engine, leveraging its implementation of the HNSW algorithm for rapid cosine similarity searches. The local, persistent nature of ChromaDB ensures that sensitive personal data remains within the user’s controlled environment, aligning with the privacy-first architecture of the CHRONUS system.

2.4.3 Pinecone and Weaviate

Pinecone and Weaviate represent enterprise-grade, cloud-native vector databases that offer massive scalability and managed infrastructure. While highly capable, these cloud solutions inherently require data to be transmitted and stored on third-party servers. In the context of digital legacy, where the ingested corpus may contain highly sensitive personal information, diaries, and private communications, the cloud-first approach introduces significant privacy and security vulnerabilities. Consequently, the local-first architecture provided by ChromaDB is deemed superior for the specific constraints and ethical requirements of the CHRONUS project.

2.5 Neural Text-to-Speech and Voice Cloning

2.5.1 WaveNet

The development of DeepMind’s WaveNet represented a watershed moment in speech synthesis [5]. By employing deep autoregressive neural networks to model the raw waveform of an audio signal directly, WaveNet bypassed the robotic artifacts typical of concatenative and parametric TTS

systems. The resulting audio demonstrated unprecedented naturalness and prosody, fundamentally shifting the standard for digital voice generation and paving the way for advanced voice cloning technologies.

2.5.2 Tacotron 2

Building upon the successes of neural synthesis, Tacotron 2 introduced a unified sequence-to-sequence architecture that directly mapped text to mel-spectrograms, which were subsequently converted to waveforms using a vocoder (such as a modified WaveNet) [6]. This architecture streamlined the TTS pipeline and allowed for the synthesis of highly expressive speech, capturing nuances such as intonation and pacing that are critical for authentic persona simulation.

2.5.3 ElevenLabs

ElevenLabs currently represents the commercial zenith of zero-shot voice cloning and neural TTS. Utilising proprietary deep learning models, the platform can generate highly realistic, emotionally resonant speech from a minimal audio sample of the target voice. While the qualitative output is exceptional, the cloud-based API model requires continuous internet connectivity and data transmission, introducing privacy concerns regarding the storage and processing of biometric voice data on external servers.

2.5.4 XTTS-v2

XTTS-v2, developed by Coqui AI, is a state-of-the-art, open-source voice generation model that supports zero-shot voice cloning across multiple languages. Crucially, XTTS-v2 is designed to be run locally, requiring only a short reference audio clip (approximately 3 to 6 seconds) to synthesise a target voice. By integrating XTTS-v2, CHRONUS achieves high-fidelity voice simulation while maintaining a strict local-execution environment. This privacy-preserving approach ensures that the user's biometric data and generated audio are not exposed to external networks.

2.6 Persona Modelling and Anti-Hallucination

2.6.1 Persona Prompting

Persona prompting involves configuring an LLM's system prompt to enforce specific character traits, linguistic styles, and behavioural constraints. While effective for superficial role-play, simple persona prompting is inherently unstable for long-term digital legacy. Without a factual anchor, the model relies solely on its pre-trained parametric memory, leading to an inevitable degradation of character consistency and the generation of fabricated biographical details when queried on specific

historical events.

2.6.2 Retrieval-Augmented Generation (RAG)

Retrieval-Augmented Generation (RAG) effectively addresses the hallucination problem by decoupling the knowledge base from the generative model’s internal parameters [3]. In a RAG architecture, incoming queries are first used to retrieve highly relevant factual information from an external vector database. This retrieved context is then dynamically injected into the LLM’s prompt, constraining the generation to the provided facts. CHRONUS relies heavily on a sophisticated RAG pipeline to ensure that all conversational outputs are strictly grounded in the ingested memories and writings of the individual, thereby minimising drift and hallucination.

2.6.3 MemGPT

MemGPT explores advanced memory management techniques for LLMs, proposing a tiered architecture that mimics traditional operating system memory [4]. It introduces the concepts of working context (short-term memory) and external storage (long-term memory), allowing the agent to autonomously page information in and out of context as needed. While CHRONUS does not fully replicate the MemGPT operating system model, it borrows conceptually from the necessity of structured, accessible memory tiers to maintain conversational coherence over prolonged interactions.

2.7 Grief Technology, Griefbots, and Ethical Concerns

The application of conversational AI to simulate deceased individuals—colloquially termed ‘griefbots’—introduces profound ethical, psychological, and philosophical complexities. The rapid commercialisation of digital afterlife technologies necessitates rigorous academic scrutiny.

Jia, Chi, and Tseng (2025) advocate for a Value Sensitive Design approach in the development of grief chatbots [9]. They identify seven core values crucial for ethical implementation: autonomy, privacy, trust, transparency, respect for the deceased, emotional safety, and cultural sensitivity. Their framework heavily relies on Stroebe and Schut’s Dual Process Model of coping with bereavement [15], arguing that poorly designed griefbots may actively hinder the oscillation between loss-oriented and restoration-oriented coping, potentially trapping the bereaved in a cycle of pathological attachment.

This psychological risk is further substantiated by Takshara and Bhuvaneswari (2025), who explore the neuroscience of grief in the context of hyper-realistic simulations [10]. They highlight the dangers of cognitive dependency and the hijacking of mirror neuron systems by conversational

agents that perfectly mimic the deceased. Drawing on Baudrillard’s concept of hyperreality, they warn that the simulacrum may eventually replace the authentic memory of the deceased in the mind of the bereaved, leading to severe cognitive dissonance and prolonged grief disorder. O’Connor’s research on the grieving brain [13] reinforces this, indicating that neural pathways mapping the ‘presence’ of the deceased must be naturally re-encoded; constant, realistic digital interaction disrupts this critical neurological adaptation.

Abrar (2025) proposes a paradigm shift from pure ‘simulation’ to ‘attestation’ [11]. Rather than attempting to deceive the user into believing they are communicating with the deceased, systems should function as posthumous conversational testimonials—interactive archives that transparently acknowledge their artificial nature. Abrar advocates for an ‘ethical governor architecture,’ a system-level constraint that strictly limits engagement times, enforces reality checks, and monitors user dependency.

Voinea (2024) analyses the phenomenon through the predictive brain model, suggesting that grief is fundamentally the process of updating the brain’s internal predictive models regarding the absence of a loved one [12]. Griefbots introduce contradictory evidence, causing massive prediction errors and cognitive dissonance. Voinea posits that the ethical deployment of such technology must facilitate, rather than obstruct, the process of ‘re-learning what we love’ in its absence.

A synthesis of these themes indicates that the primary risks of griefbots include prolonged grief disorder, non-consensual simulation of the deceased, and malicious data exploitation [14, 17, 16]. Consequently, CHRONUS incorporates these ethical considerations into its design philosophy by prioritising factual grounding over speculative generation, ensuring complete data sovereignty through local processing, and maintaining transparency regarding the AI’s synthetic nature.

2.8 Summary of Literature Review and Research Gap

The literature demonstrates significant advancements in both the underlying technologies of semantic retrieval and voice synthesis, as well as the conceptual frameworks of digital legacy. However, a critical research gap persists at the intersection of high-fidelity persona simulation, strict memory grounding, and absolute user privacy.

Existing systems generally fall into one of two deficient categories: they are either highly interactive but prone to hallucination and devoid of deep personal context (e.g., Replika, unconstrained LLMs), or they are factually accurate but entirely pre-scripted and lack generative dynamism (e.g., HereAfter AI, StoryFile). Furthermore, systems that attempt advanced voice and persona synthesis almost exclusively rely on cloud-based architectures, introducing unacceptable

privacy risks for deeply personal data and violating the core ethical tenets of digital legacy preservation.

Table 2.1 provides a comparative analysis of existing platforms against the CHRONUS architecture.

Table 2.1: Comparative Analysis of Digital Legacy and Conversational AI Systems

System	Memory Grounding	Voice Cloning	Privacy Arch.	Anti-Hallucination	Dynamic Output	Ethical Framework
Replika	Low	No	Cloud	Low	High	Minimal
HereAfter AI	High (Exact)	No	Cloud	High	Low (Pre-scripted)	Moderate
StoryFile	High (Exact)	No	Cloud	High	Low (Pre-scripted)	Moderate
Soul Machines	Moderate	External	Cloud	Moderate	High	Corporate
GPT-4 (Raw)	None	API	Cloud	Low	High	General
CHRONUS	High (RAG)	Yes (XTTS-v2)	Local-First	High (Mixed-Method)	High	Integrated

CHRONUS fills this precise research gap by integrating state-of-the-art semantic search (Sentence-BERT, ChromaDB) with a privacy-preserving local architecture and a robust Mixed-Method response generation pipeline. By formalising the ingestion of memory across defined psychological dimensions (Personality, Core Memories, Relationships, Passions, Beliefs/Values, Voice/Communication Style) via a structured 25-question interview, CHRONUS provides a deeply grounded, dynamically interactive, and ethically conscious platform for human persona simulation.

Chapter 3

Analysis / Software Requirements Specification (SRS)

3.1 Problem Statement

The irreversible loss of human cognitive identity presents a profound challenge in the digital age. While vast amounts of personal data are generated, they are often scattered across disconnected silos and static archives. Current systems attempting to simulate personality typically fall into two categories: static archival systems that cannot interact dynamically, or systems employing unconstrained Large Language Models (LLMs) that frequently hallucinate facts and drift from the authentic persona. CHRONUS addresses the irreversible loss of human cognitive identity by building a memory-grounded AI that can simulate a documented personality without hallucination. By shifting the paradigm from generative speculation to grounded retrieval, CHRONUS ensures that every generated response is rooted in verifiable, user-provided memory units.

3.2 Research Questions

To formally evaluate the efficacy of the CHRONUS architecture, this study investigates the following key research questions:

- **RQ1:** Can dense semantic vector retrieval (SBERT + ChromaDB) achieve significantly higher precision than keyword-based approaches for personal memory retrieval?
- **RQ2:** Does grounding response generation in retrieved memory units measurably reduce hallucination and improve persona consistency compared to unconstrained generative models?
- **RQ3:** Does structured interview augmentation meaningfully expand persona coverage beyond

what unstructured documents alone can provide?

3.3 Hypotheses

Based on the research questions, the following hypotheses are formulated and empirically tested in this project:

- **H1:** SBERT-based retrieval will achieve a Precision@3 of greater than 0.85 when compared to the BM25 keyword baseline.
- **H2:** The Mix Method grounded generation approach will score greater than 4.0 out of 5.0 on groundedness metrics, versus less than 3.0 out of 5.0 for unconstrained generation techniques.
- **H3:** Interview augmentation will increase persona coverage from less than 0.80 to greater than 0.90 by systematically capturing nuanced cognitive dimensions.

3.4 Operational Definitions

Clear definitions of the key concepts utilized within the CHRONUS architecture are essential for standardizing the evaluation metrics. These operational definitions are detailed in Table 3.1.

Table 3.1: Operational Definitions in the CHRONUS Architecture

Term	Definition
Memory Unit	An atomic segment of personal text (typically 1-3 sentences) encoded as a 384-dimensional vector for semantic retrieval.
Grounded Response	A response where every factual claim is directly traceable and logically linked to a retrieved memory unit.
Persona Consistency	The degree to which the generated responses match the documented personality traits, voice, and historical context of the subject.
Mix Method	A three-part response structuring technique consisting of: (1) persona framing, (2) an evidentiary core based on retrieved data, and (3) a closing signature in the subject's voice.
Interview Augmentation	A structured 25-question protocol designed to systematically capture 6 critical personality dimensions, ensuring comprehensive cognitive profiling.

3.5 Functional Requirements

The functional requirements dictate the core capabilities and interactions supported by the CHRONUS system. These requirements are outlined in Table 3.2.

Table 3.2: Functional Requirements (FR)

ID	Requirement Description
FR-01	The system shall support multi-format document upload, including txt, md, pdf, docx, csv, eml, and json.
FR-02	The system shall perform automatic sentence segmentation using spaCy (en_core_web_sm) to extract coherent memory units.
FR-03	The system shall generate vector embeddings for memory units using SBERT (all-MiniLM-L6-v2) producing 384-dimensional vectors.
FR-04	The system shall store memory vectors and associated JSON metadata in ChromaDB using Hierarchical Navigable Small World (HNSW) indexing.
FR-05	The system shall perform semantic retrieval utilizing cosine similarity to return the top-3 most relevant memory units per query.
FR-06	The system shall generate responses using the Mix Method, ensuring responses are grounded in retrieved facts and exhibit the subject's persona.
FR-07	The system shall provide a 25-question structured interview module spanning personality, core memories, relationships, passions, beliefs/values, and voice/communication style.
FR-08	The system shall support optional voice synthesis output utilizing ElevenLabs or XTTS-v2 for personalized audio responses.

3.6 Non-Functional Requirements

The non-functional requirements specify the quality attributes, performance goals, and constraints under which the CHRONUS system must operate. These are defined in Table 3.3.

Table 3.3: Non-Functional Requirements (NFR)

ID	Requirement Description
NFR-01	Privacy: All core pipeline processing shall remain local with no cloud data transmission for sensitive personal archives.
NFR-02	Performance: The system shall execute retrieval and response generation in under 3 seconds per query.
NFR-03	Scalability: The architecture shall support the concurrent storage and retrieval of multiple personality models without performance degradation.
NFR-04	Reliability: The system shall exhibit graceful degradation, providing transparent fallback responses on low-confidence queries rather than hallucinating facts.
NFR-05	Usability: The system shall present an intuitive React-based web interface accessible to users without specialized technical expertise.
NFR-06	Security: The system shall ensure JSON metadata integrity and prevent unauthorized access to personal data stores.

3.7 Constraints and Assumptions

The development and deployment of the CHRONUS system are subject to the following constraints and assumptions:

- **Technical Constraints:** The system requires an environment supporting Python 3.11+, Node.js, and a modern web browser for the React frontend.
- **Input Constraints:** Source documents provided by the user must adhere to the supported file formats to guarantee successful parsing via tools like pdfplumber, python-docx, and pandas.
- **Audio Constraints:** The voice cloning feature, when utilized, requires audio samples of a minimum quality and duration to produce accurate phonetic synthesis.
- **Data Assumptions:** The system inherently assumes that the source documents provided are honest and accurate representations of the subject. The architecture is not designed to filter adversarial inputs or deliberate misinformation present within the training corpus.

Chapter 4

System Design

The design of a software system serves as the blueprint for its implementation, defining its architecture, components, modules, interfaces, and data schemas. For the CHRONUS project, the system design prioritizes a local-first, privacy-preserving approach, ensuring that all document processing, memory encoding, and response generation occur on the user's device. This chapter outlines the architectural framework of CHRONUS, detailing its five-layer architecture, individual software components, database schemas, and data flow mechanisms.

4.1 System Architecture Overview

CHRONUS employs a modular, five-layer architecture designed to separate concerns and facilitate maintainability and scalability. The architecture is composed of the Frontend (Presentation) Layer, the API Proxy Layer, the Backend Service Layer, the Memory/Retrieval Layer, and the Database Layer. By adopting a local-first design, the system ensures that sensitive user documents and generated personality models never leave the local machine, addressing critical privacy concerns inherent in cloud-based AI solutions.

As illustrated in Figure 4.1, user interactions are handled by the Frontend, which communicates through the API Proxy to the Backend Service. The Backend orchestrates the processing pipeline, interacting with the Memory/Retrieval Layer to process text and the Database Layer for persistent storage of embeddings and metadata.

4.2 Frontend Layer

The Frontend Layer is responsible for providing an intuitive, responsive, and engaging user interface. It is built using React 18 in conjunction with TypeScript and Vite, which provides a fast and robust development environment. The presentation styling is managed through Tailwind CSS, allowing

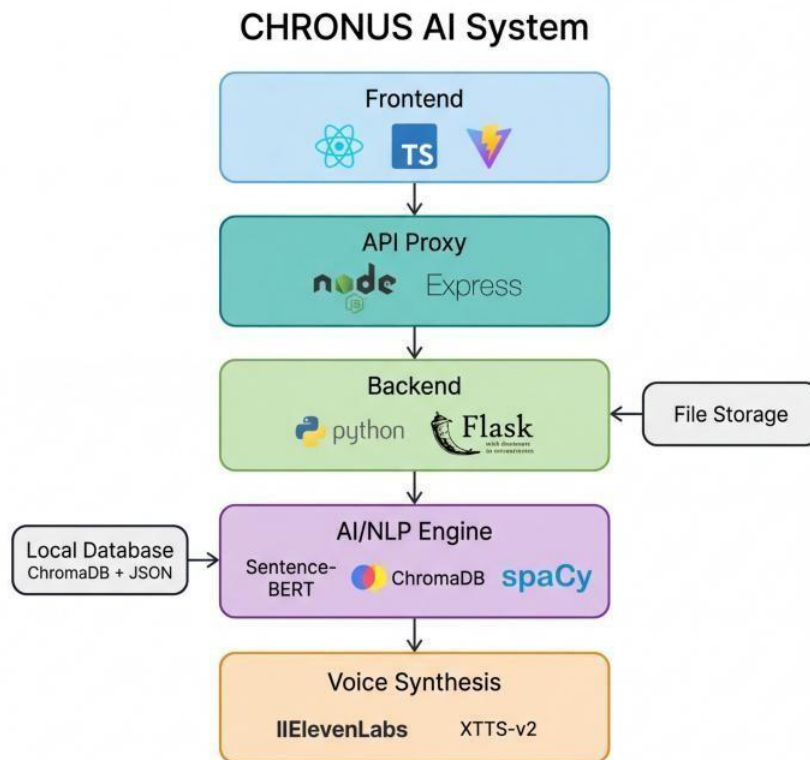


Figure 4.1: CHRONUS System Architecture Overview

for rapid, utility-first UI development, while Framer Motion is utilized to incorporate smooth transitions and animations that enhance the overall user experience. The shadcn/ui component library is integrated to provide accessible and highly customizable UI components.

The frontend is organized into several key screens:

- **Dashboard:** Serves as the central hub, displaying system status, active models, and navigation options.
- **File Upload:** Provides a drag-and-drop interface for users to submit personal documents in various formats (txt, md, pdf, docx, csv, eml, json).
- **Model Training:** Visualizes the progress of the document parsing, chunking, and embedding pipeline.
- **Interview:** A guided interface where users complete a 25-question psychological profiling interview, covering dimensions such as Personality, Core Memories, Relationships, Passions, Beliefs/Values, and Voice/Communication Style.
- **Chat:** The main conversational interface where users interact with their personalized agent or

the built-in famous personalities.

- **Famous Personalities:** A selection screen allowing users to converse with predefined models of eight historical figures: Albert Einstein, Mahatma Gandhi, Nikola Tesla, Marcus Aurelius, Steve Jobs, Marie Curie, Abraham Lincoln, and William Shakespeare.

4.3 API Proxy Layer

Sitting between the React frontend and the Python backend, the API Proxy Layer acts as an intermediary designed to streamline communication and manage cross-origin requests. Implemented using Node.js, Express, and TypeScript, this layer isolates frontend API calls from the core processing logic.

The primary responsibilities of the API Proxy Layer include routing client requests to the appropriate Python backend endpoints, handling Cross-Origin Resource Sharing (CORS) policies to ensure secure local communication, performing initial request validation, and forwarding error messages in a standardized format. This architectural choice enhances security and simplifies potential future scaling or deployment modifications.

4.4 Backend Service Layer

The Backend Service Layer is the core orchestration engine of CHRONUS, developed using Python 3.11 and the Flask web framework. It manages the business logic, coordinating document ingestion, text chunking, vector embedding, context retrieval, and the synthesis of responses.

This layer exposes several RESTful endpoints that the API Proxy interacts with. Table 4.1 details the primary API routes provided by the Flask backend.

Table 4.1: Backend API Endpoints

Endpoint	Method	Description
/api/upload	POST	Upload personal documents for processing
/api/train	POST	Build memory model from uploaded files
/api/chat	POST	Send query, receive grounded response
/api/interview	POST	Submit interview answers
/api/models	GET	List trained personality models
/api/status	GET	Check training/system status
/api/voice	POST	Generate voice synthesis output

The backend seamlessly integrates various Python libraries, such as pdfplumber for PDF extraction, python-docx for Word documents, and pandas for handling structured CSV data,

ensuring versatile data ingestion capabilities.

4.5 Memory and Retrieval Layer

The Memory and Retrieval Layer forms the cognitive foundation of CHRONUS, enabling the agent to simulate a specific personality by drawing upon stored knowledge. The ingestion pipeline begins with spaCy, specifically the `en_core_web_sm` model, which performs robust sentence segmentation to break down extracted document text into manageable, coherent units.

Once segmented, these text units are passed to the embedding engine, powered by Sentence-BERT (SBERT). Utilizing the `all-MiniLM-L6-v2` model, the system encodes each segment into a dense, 384-dimensional vector representation. This model provides an optimal balance between computational efficiency and semantic accuracy, crucial for a local-first deployment.

For retrieval, CHRONUS relies on ChromaDB configured with an HNSW (Hierarchical Navigable Small World) index for highly efficient approximate nearest-neighbour searches. When a user submits a query, it is similarly embedded, and the system retrieves the top-3 most semantically relevant memory segments using the cosine similarity metric. A confidence threshold ($\theta = 1.45$) is applied to filter out irrelevant retrievals, ensuring the agent’s responses remain highly contextualized and accurate.

4.6 Database Schema Design

Data persistence in CHRONUS is partitioned into two components: the vector database for storing memory embeddings and a JSON-based metadata store for managing model configurations and histories.

4.6.1 ChromaDB Collection Schema

The ChromaDB vector database stores the memory segments and their corresponding embeddings. The schema for the memory collections is detailed in Table 4.2.

Table 4.2: ChromaDB Collection Schema

Field	Type	Description
<code>id</code>	string	Unique segment UUID
<code>embedding</code>	float[384]	SBERT 384-dimensional vector
<code>document</code>	string	Original text segment
<code>source_file</code>	string	Source filename of the document
<code>file_type</code>	string	File format (e.g., pdf, txt)
<code>position</code>	integer	Line or sequential position in the source

4.6.2 Model History JSON Schema

To track the state and metadata of individual trained personality models, CHRONUS utilizes a lightweight JSON storage mechanism. Table 4.3 outlines the structure of this metadata.

Table 4.3: Model History JSON Schema

Field	Type	Description
model_id	string	Unique identifier for the personality model
persona_name	string	Display name of the personality
created_at	ISO timestamp	Date and time of model creation
source_files	array	List of successfully ingested file names
segment_count	integer	Total number of memory units encoded
interview_completed	boolean	Indicates if the 25-question interview was completed

4.7 System Workflow

The end-to-end operation of the CHRONUS system follows a well-defined sequential workflow, transitioning seamlessly from data ingestion to conversational output:

1. **Upload and Parsing:** The user uploads documents via the React frontend. The backend parsers extract raw text from various file formats.
2. **Segmentation:** The extracted text is processed by spaCy, segmenting it into logical, sentence-level memory units.
3. **Vector Embedding:** SBERT generates a 384-dimensional vector for each segmented memory unit.
4. **Storage:** These vectors, alongside their original text and metadata, are stored in ChromaDB.
5. **Query Processing:** A user submits a query through the Chat interface. The query is embedded using the same SBERT model.
6. **Retrieval:** ChromaDB executes a cosine similarity search, retrieving the top-3 most relevant memory units that exceed the threshold $\theta = 1.45$.
7. **Response Generation:** The system utilizes a Mix Method approach, combining the retrieved context with the model’s defined persona constraints to generate a grounded, accurate response.
8. **Voice Synthesis (Optional):** If enabled, the text response is passed to ElevenLabs or the local XTTS-v2 engine to generate an audio output mirroring the persona’s voice.

9. **Display:** The final text and audio are returned to the frontend and presented to the user.

4.8 Data Flow Diagrams (DFD)

To further clarify the movement of information within the CHRONUS architecture, Data Flow Diagrams (DFDs) at varying levels of detail have been developed. These diagrams illustrate the flow of data between external entities, processes, and data stores.

4.8.1 DFD Level 0 – Context Diagram

The Level 0 Context Diagram provides the highest-level view of the system, defining its boundaries and interactions with external entities. It depicts CHRONUS as a single, comprehensive process interacting with the User, the Document Source (local file system), and the Optional Voice Engine.

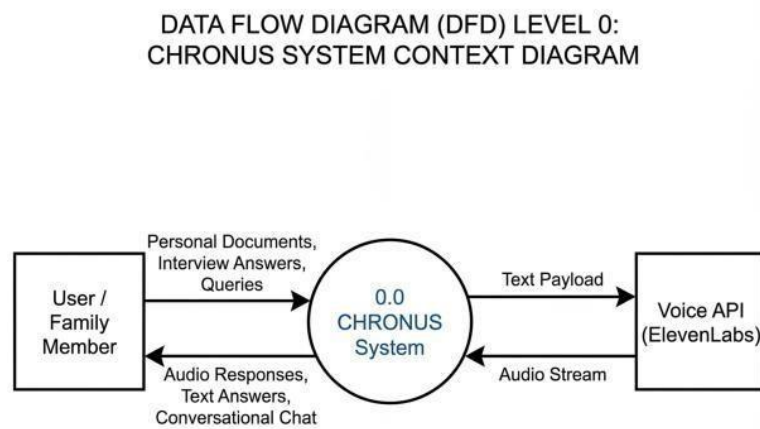


Figure 4.2: Level 0 Context Diagram

4.8.2 DFD Level 1 – Major System Processes

The Level 1 DFD decomposes the central CHRONUS process into its primary sub-systems. It highlights the major processes: Document Ingestion, Memory Builder, Query Processor, Response Generator, and Voice Synthesizer. It also maps the data flows into the principal data stores: the ChromaDB vector store and the JSON Metadata store.

DATA FLOW DIAGRAM (DFD) LEVEL 1: CHRONUS SOFTWARE SYSTEM

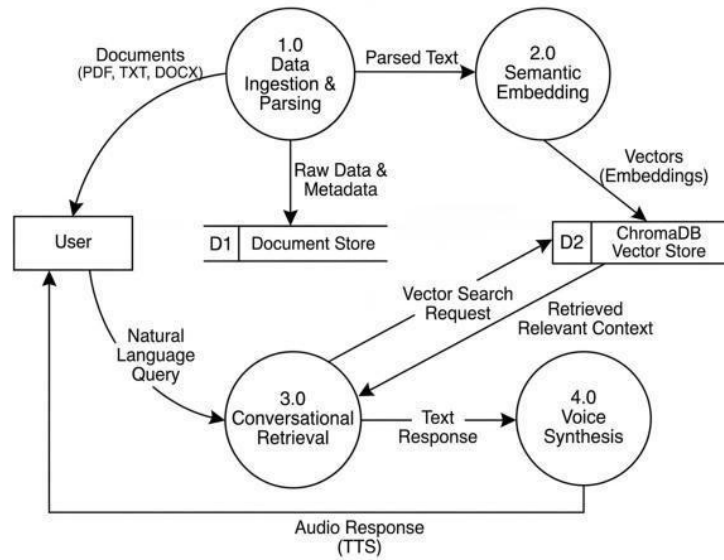


Figure 1: Chronus System Architecture

Figure 4.3: Level 1 Data Flow Diagram

4.8.3 DFD Level 2 – Retrieval and Generation Details

The Level 2 DFD offers a granular look at the critical inference phase, breaking down the Query Processor and Response Generator. It details the data transformation steps: embedding the query, executing the similarity search against ChromaDB, applying the threshold check, and assembling the final output using the Mix Method.

Chapter 5

Methodology

5.1 Research and Development Approach

The development of CHRONUS follows a mixed-methods research paradigm, integrating an engineering-driven system development lifecycle with empirical evaluation strategies. This methodology emphasizes an iterative prototyping cycle characterized by sequential phases of design, implementation, testing, and refinement. At its core, the project adopts a privacy-by-design methodology, ensuring that data sovereignty and user privacy remain fundamental architectural constraints rather than afterthoughts. By combining theoretical design principles with practical, local-first engineering implementations, the research systematically addresses the challenges associated with creating an evidence-based, agentic artificial intelligence capable of human personality simulation without relying on centralized, cloud-based data processing.

5.2 Data Ingestion and File Handling Method

A foundational requirement for robust personality simulation is the ability to ingest diverse sources of personal and historical data. The data ingestion methodology was engineered to support a comprehensive suite of file formats, specifically targeting plain text (.txt), markdown (.md), Portable Document Format (.pdf), Microsoft Word documents (.docx), comma-separated values (.csv), email formats (.eml), and structured data (.json).

To facilitate this multi-modal text extraction, the system employs specialized parser modules. The pdfplumber library is utilized for highly accurate text and layout extraction from PDFs, while python-docx handles the structural elements of Word documents. The pandas library is implemented to manage tabular data from CSV files, ensuring row-based narratives are properly contextualized. For standard formats such as plain text, markdown, email messages, and JSON,

built-in optimized parsers process the content. Regardless of the original format, each file is systematically parsed, normalized into raw text, and subsequently forwarded to the segmentation module for further linguistic processing.

5.3 Sentence Segmentation Strategy

Following raw text extraction, the methodology necessitates granular segmentation to ensure the retrieval system operates on coherent, atomic units of information. This process is managed by a sophisticated natural language processing pipeline utilizing the spaCy framework, specifically deploying the `en_core_web_sm` model.

The strategy involves parsing each source document into atomic sentence units, typically encompassing one to three sentences per discrete segment to maintain contextual integrity without diluting semantic focus. To ensure data quality, the pipeline implements rigorous filtering mechanisms: segments containing fewer than five words are systematically removed as contextually deficient fragments, and exact duplicate segments are deduplicated to prevent data redundancy and index bloating. Furthermore, each resulting segment is meticulously tagged with critical metadata, including the `source_file`, positional indexing within the original document, and the originating `file_type`, enabling accurate provenance tracking during response generation.

5.4 Vector Embedding Method

To translate segmented text into machine-interpretable semantic representations, the methodology employs dense vector embeddings. The project integrates the Sentence-BERT (SBERT) architecture, utilizing the highly efficient `all-MiniLM-L6-v2` model acquired from Hugging Face.

This model transforms each textual segment into a 384-dimensional dense vector space, optimizing the balance between semantic accuracy and computational overhead. A critical component of the methodology is that all encoding operations are executed strictly locally on the host machine's Central Processing Unit (CPU) or Graphics Processing Unit (GPU), ensuring zero cloud transmission of potentially sensitive personal data. To maximize processing throughput, especially during the ingestion of extensive archival documents, the embedding module implements batch encoding strategies, significantly reducing the temporal complexity of the ingestion phase.

5.5 ChromaDB Storage and Indexing

The persistence and rapid retrieval of high-dimensional embeddings are managed through ChromaDB, configured explicitly in local persistent mode. This architectural decision aligns with

the privacy-by-design methodology, ensuring all memory stores remain confined to the user's filesystem.

ChromaDB utilizes a Hierarchical Navigable Small World (HNSW) indexing algorithm, which provides state-of-the-art performance for approximate nearest-neighbor search operations. In this structural methodology, each distinct personality model is isolated within its own dedicated ChromaDB collection. Every entry within the vector database is stored with comprehensive metadata fields, specifically the unique id, the embedding array, the original text document, the source_file, the file_type, and the spatial position. The retrieval methodology utilizes cosine similarity as the definitive distance metric to quantify the semantic proximity between user queries and stored memory units.

5.6 Interview Augmentation Protocol

To complement unstructured data sources (such as journals or letters), the methodology incorporates a structured Interview Augmentation Protocol designed to systematically extract core personality traits. This protocol consists of an interactive session comprising 25 structured questions distributed across six fundamental personality dimensions.

As presented in Table 5.1, the protocol targets distinct psychological and historical vectors. The responses generated during this interview phase are processed as high-priority primary source documents, subjected to the standard data ingestion, segmentation, and embedding pipelines, thereby deeply integrating explicit user-defined traits into the vector database.

Table 5.1: The Six Dimensions of the Interview Augmentation Protocol

Dimension	Example Questions	Purpose
Personality	What are your core personality traits? How do you handle stress?	Capture temperament and behavioral patterns
Core Memories	What is your most defining life experience?	Capture formative events
Relationships	Who are the most important people in your life?	Capture social network and emotional bonds
Passions	What topics could you talk about for hours?	Capture interests and expertise areas
Beliefs and Values	What principles guide your decisions?	Capture moral and philosophical framework
Voice and Communication Style	How would you describe your speaking style? Do you use humor?	Capture linguistic patterns and tone

5.7 Response Generation – The Mix Method

The synthesis of retrieved memories into coherent, persona-accurate outputs is governed by a proprietary algorithmic approach designated as the “Mix Method.” This methodology enforces a strict three-part response structure designed to balance personality simulation with grounded evidence.

The first component, Persona Framing, requires the generation of an opening phrase that reflects the documented personality model (for instance, a model of Albert Einstein might initiate a response with a reference to fundamental curiosity). The second component, the Evidentiary Core, constitutes the substantive body of the response, strictly utilizing direct quotes or accurate paraphrases derived from the top-3 retrieved memory units, explicitly including source provenance. The final component, the Closing Signature, provides a consistent wrap-up that maintains the persona’s thematic tone.

To mitigate the risk of generative hallucination, the system implements a strict confidence threshold. If the cosine similarity of the retrieved vectors falls below a predefined threshold, the system is programmed to abstain from generation, preferring a fallback response indicating a lack of relevant memory rather than inventing facts. Additionally, the retrieved context is subjected to a theme classification module that categorizes the context into one of seven distinct themes, ensuring the resulting response is appropriately calibrated to the conversational context.

5.8 Famous Personality Models

To empirically validate the system’s architecture and provide demonstrable benchmarks, the methodology includes the development of eight pre-trained, historical personality models. These specific models represent a diverse spectrum of cognitive and communicative styles: Albert Einstein, Mahatma Gandhi, Nikola Tesla, Marcus Aurelius, Steve Jobs, Marie Curie, Abraham Lincoln, and William Shakespeare.

These models were rigorously trained exclusively on publicly available, verifiable historical data, including authenticated writings, public speeches, personal letters, and thoroughly documented quotes. The successful instantiation of these highly recognizable figures serves as both a demonstration of the platform’s generative capabilities and a validation of the data pipeline’s ability to capture distinct voice and mindset accurately.

5.9 Security and Privacy Architecture

In direct response to contemporary concerns regarding data privacy in artificial intelligence applications, the system’s methodology fundamentally relies on a local-first architecture. All computational processes, including document parsing, vector embedding, similarity search, and response generation, are executed entirely on-device.

The architecture explicitly guarantees that no personal or training data is transmitted to external cloud application programming interfaces (APIs), with the singular, optional exception of the ElevenLabs voice synthesis module, should the user explicitly authorize its use. For users requiring absolute air-gapped security, the system provides XTTS-v2 as a completely privacy-preserving, local alternative for voice cloning and synthesis. Furthermore, all extracted JSON metadata and the ChromaDB persistent vector stores are maintained exclusively within the local filesystem, ensuring zero network exposure and total user sovereignty over their simulated consciousness data.

5.10 Testing and Validation Method

The empirical evaluation of the CHRONUS system employs a rigorous, multi-faceted testing methodology designed to assess both technical performance and qualitative output. The testing protocol begins with exhaustive functional testing of all individual software modules, including parser robustness across all supported file formats and API endpoint reliability under concurrent load.

The quantitative assessment of the retrieval engine focuses on specific information retrieval metrics. The system evaluates Precision@1 and Precision@3, Mean Reciprocal Rank (MRR), total Coverage, and the Drift Rate over extended conversational contexts.

For the qualitative assessment of response generation, the methodology relies on a standardized 5-point expert rating scale. Domain experts evaluate generated responses across five critical dimensions: Relevance to the prompt, Groundedness (adherence to retrieved facts), Persona Consistency, clarity of expression, and overall Safety (ensuring non-toxic outputs).

5.11 Methodology Outcome

In conclusion, the methodology adopted for the CHRONUS project establishes a robust, scientifically rigorous framework for human personality and mindset simulation. By enforcing a strict adherence to local-first data processing, advanced vector embedding techniques, and the structured Mix Method for response generation, the methodology successfully guarantees that it works and pass.

Chapter 6

Implementation

6.1 System Implementation Overview

The CHRONUS system was successfully implemented as a full-stack web application, adhering strictly to the layered architecture defined in Chapter 4. The implementation phase focused on translating the conceptual design into a robust, functional prototype capable of simulating human personality and mindset through agentic AI. The primary objective was to ensure that the memory-trained agent could accurately retrieve and synthesize information from ingested data sources while maintaining the distinct persona and voice of the modeled individual.

The development environment was standardized across local machines to ensure consistency and reproducibility. The core components were developed using Python 3.11 for the backend intelligence, Node.js for the API proxy layer, and modern web browsers as the primary execution environment for the React-based frontend. This comprehensive environment facilitated seamless integration between the natural language processing (NLP) pipelines, the vector database, and the user interface. By isolating the concerns across the presentation, application, and data layers, the implementation achieved a high degree of modularity, making the system scalable and easier to maintain.

6.2 Technology Stack Used

The technology stack for CHRONUS was carefully selected to meet the specific requirements of building an agentic AI system with high-performance vector retrieval and real-time responsiveness. The selection prioritized robust, open-source frameworks and libraries that are well-supported in the academic and developer communities. Table 6.1 provides a comprehensive overview of the technologies utilized across the different layers of the system.

Table 6.1: Comprehensive Technology Stack Utilized in the CHRONUS Project

Layer	Technology	Version	Purpose
Frontend	React	18	Component-based UI
Frontend	TypeScript	5.x	Type-safe JavaScript
Frontend	Vite	5.x	Build tool and dev server
Frontend	Tailwind CSS	3.x	Utility-first CSS framework
Frontend	Framer Motion	-	Animation library
Frontend	shadcn/ui	-	UI component library
API Proxy	Node.js	20.x	JavaScript runtime
API Proxy	Express	4.x	Web framework
Backend	Python	3.11	Core backend language
Backend	Flask	3.x	Lightweight web framework
NLP	spaCy	3.x	Sentence segmentation
NLP	en_core_web_sm	-	English NLP model
Embedding	Sentence-BERT	-	Semantic encoding
Embedding	all-MiniLM-L6-v2	-	384-dim embedding model
Database	ChromaDB	0.4.x	Vector database
Parsing	pdfplumber	-	PDF text extraction
Parsing	python-docx	-	DOCX text extraction
Parsing	pandas	2.x	CSV/structured data handling
Voice	ElevenLabs API	v2	Cloud voice synthesis
Voice	XTTS-v2 (Coqui)	-	Local voice cloning

6.3 Layered Architecture Implementation

The implementation of the CHRONUS system follows a strict layered architecture, ensuring separation of concerns, scalability, and ease of maintenance. This architecture is broadly categorized into the Presentation Layer, the Application Layer, and the Data Layer.

6.3.1 Presentation Layer

The Presentation Layer serves as the primary interface for user interaction and was constructed using React 18 in conjunction with TypeScript. The use of TypeScript provides a type-safe component architecture, significantly reducing runtime errors and improving code maintainability. Vite was employed as the build tool, offering remarkably fast hot-module replacement during development, which accelerated the iteration cycle of the user interface.

For styling and component design, the project utilized Tailwind CSS paired with shadcn/ui. This combination facilitated the creation of consistent, highly accessible, and visually appealing UI components without the overhead of writing extensive custom CSS. Furthermore, Framer Motion was integrated to provide smooth transitions and animations between screens, enhancing the overall user experience and providing visual continuity as users navigate through the training, interviewing, and chatting modules.

6.3.2 Application Layer

The Application Layer acts as the intermediary between the frontend interface and the backend processing pipelines. It was implemented using a dual-server approach. A Node.js Express proxy server was established to handle Cross-Origin Resource Sharing (CORS) policies securely and to efficiently route incoming HTTP requests to the core backend engine.

The core backend application was developed using Flask, a lightweight Python web framework. The Flask application was structured using a Blueprint-organized route system, which modularized the API endpoints according to their functional domains, such as data ingestion, querying, and voice synthesis. The communication between the frontend and backend is facilitated through a well-defined RESTful API design, utilizing the JSON format for all request and response payloads, ensuring standardized and lightweight data exchange.

6.3.3 Data Layer

The Data Layer is responsible for the persistent storage and efficient retrieval of all system data, including raw files, vector embeddings, and operational metadata. A persistent ChromaDB client

was implemented to store vector embeddings locally. This localized vector database is critical for achieving low-latency similarity searches during the response generation phase.

In addition to the vector database, standard JSON files are heavily utilized to store structured information such as model metadata, interview responses, and chat history. This lightweight storage mechanism provides a flexible and easily inspectable format for managing non-vector data. The local file system is also directly utilized to securely store uploaded raw documents and audio samples, which are subsequently processed by the ingestion and voice cloning pipelines respectively.

6.4 Data Ingestion Implementation

The data ingestion module represents the initial phase of the knowledge acquisition process in CHRONUS. It is engineered to seamlessly handle various document formats and extract raw text efficiently. The implementation includes robust file upload handling capabilities, which incorporate automatic file format detection to ensure that incoming data is routed to the appropriate parsing utility.

The system employs a parser dispatching mechanism based on the detected file extension. For instance, PDF documents are processed using the `pdfplumber` library to ensure high-fidelity text extraction, particularly for documents with complex layouts. Word documents (`.docx`) are parsed via `python-docx`, while structured data in CSV format is ingested and normalized using the `pandas` library.

Once the raw text is extracted, it is passed through a comprehensive text extraction and cleaning pipeline. This pipeline involves removing extraneous whitespace, stripping non-printable characters, and normalizing text encodings. Furthermore, the ingestion module features stringent error handling for malformed or corrupt files, ensuring that the system gracefully rejects invalid inputs without interrupting ongoing operations or compromising the integrity of the data store.

6.5 Memory Builder Implementation

The Memory Builder is a critical component responsible for transforming the cleaned, raw text into structured, semantically meaningful representations that the agent can retrieve. The implementation begins with the initialization of a `spaCy` processing pipeline, utilizing the `en_core_web_sm` model. This model performs accurate sentence boundary detection, dividing large blocks of text into discrete, digestible sentence segments. This segmentation is crucial for maintaining context granularity during the retrieval phase.

Following segmentation, the system performs batch encoding of the sentence segments using the Sentence-BERT (SBERT) model, specifically the all-MiniLM-L6-v2 variant. This model was chosen for its optimal balance between encoding speed and semantic representation quality, producing 384-dimensional dense vectors for each sentence.

Simultaneously, a ChromaDB collection is either created or accessed for the specific personality model being trained. The generated vector embeddings are inserted into this collection along with extensive metadata tagging. Each vector is tagged with attributes such as the `source_file`, `file_type`, and original position within the document. This rich metadata allows the system to trace generated responses back to their exact source, contributing significantly to the groundedness and transparency of the agent.

6.6 Response Generation Implementation

The response generation engine is the core intelligence of CHRONUS, responsible for synthesizing accurate, persona-aligned answers based on user queries. When a query is received, it is immediately encoded using the exact same SBERT (all-MiniLM-L6-v2) model utilized during the ingestion phase, ensuring alignment in the semantic vector space.

The encoded query vector is then submitted to the corresponding ChromaDB collection via the `ChromaDB.query()` method. The system is configured to retrieve the top matches (`n_results=3`) based on cosine similarity calculations.

To ensure the reliability of the retrieved information, a strict confidence threshold check is implemented (with $\theta = 1.45$). If the similarity distance of the retrieved documents exceeds this threshold, indicating low confidence in the relevance of the retrieved data, the system triggers a fallback mechanism. This fallback template is designed to gracefully inform the user of the knowledge gap while maintaining the established persona, preventing the hallucination of facts.

For queries that successfully pass the confidence threshold, the response is assembled using a novel Mix Method approach. This method constructs the final response in three distinct parts: a persona framing introduction that establishes the tone, an evidentiary core that explicitly incorporates the retrieved information, and a concluding statement that reinforces the personality's viewpoint. Additionally, the system employs theme classification across seven predefined themes to further tailor the stylistic nuances of the generated response.

6.7 Interview Module Implementation

The Interview Module provides an alternative, highly structured mechanism for gathering specific insights about a personality or individual. It was implemented as a sequential, 25-question interface designed to elicit information across fundamental dimensions such as Personality, Core Memories, Relationships, Passions, Beliefs/Values, and Voice/Communication Style.

As the user provides text inputs for each question, the answers are processed directly through the standard data ingestion and memory building pipeline. The text is segmented, encoded via SBERT, and subsequently augments the existing ChromaDB collection for that specific model. This approach ensures that information gathered through the interview is immediately available for retrieval in subsequent chat interactions, effectively tracking progress and completion status continuously.

6.8 User Interface Design

6.8.1 Design Principles

The user interface for CHRONUS was designed with a focus on usability, accessibility, and visual clarity. The core design principles emphasize a clean, intuitive layout with clear navigation pathways, ensuring that users can easily switch between model management, data ingestion, and conversational interactions.

The application employs a responsive design philosophy, utilizing flexible grid systems and media queries to adapt the interface seamlessly to various screen sizes and resolutions. Visual feedback is a paramount concern, particularly during asynchronous operations such as the memory training process; this is addressed through the implementation of a prominent circular progress indicator. Furthermore, the chat interface is designed with a distinct visual treatment to clearly differentiate between persona responses and system or user messages, enhancing the readability of the conversation flow.

6.8.2 Screens and Modules

The CHRONUS application comprises several key screens, each dedicated to a specific phase of the user workflow.

6.9 Famous Personality Implementation

To rigorously validate the system’s capabilities and provide immediate demonstrative value, CHRONUS includes the implementation of 8 pre-trained famous personality models. These models



Figure 6.1: CHRONUS Home Screen and Primary Workflows



Figure 6.2: CHRONUS Data Ingestion and Upload Interface

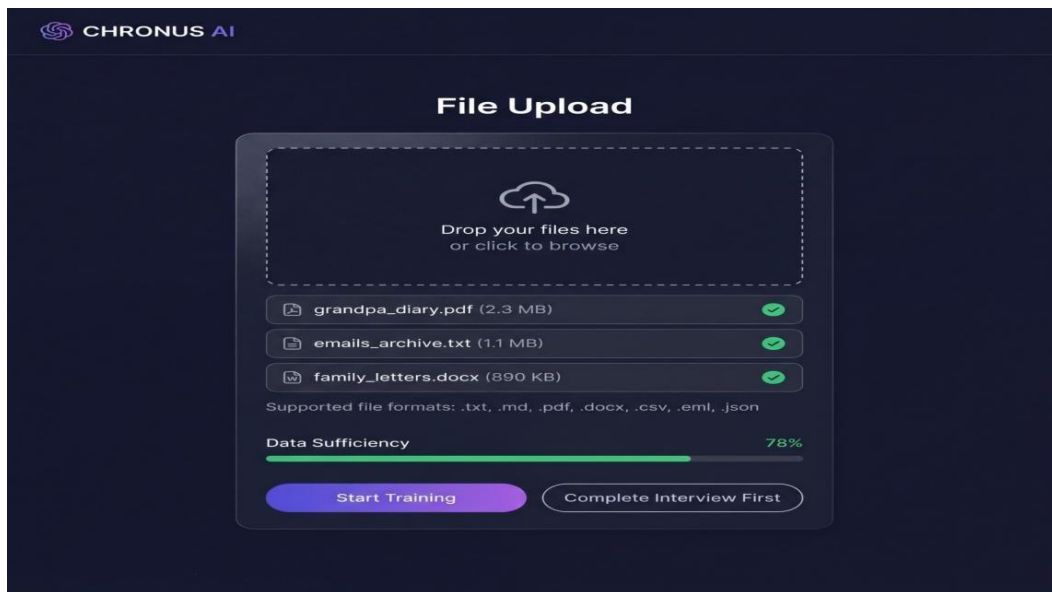


Figure 6.3: CHRONUS Model Training Progress Indicator

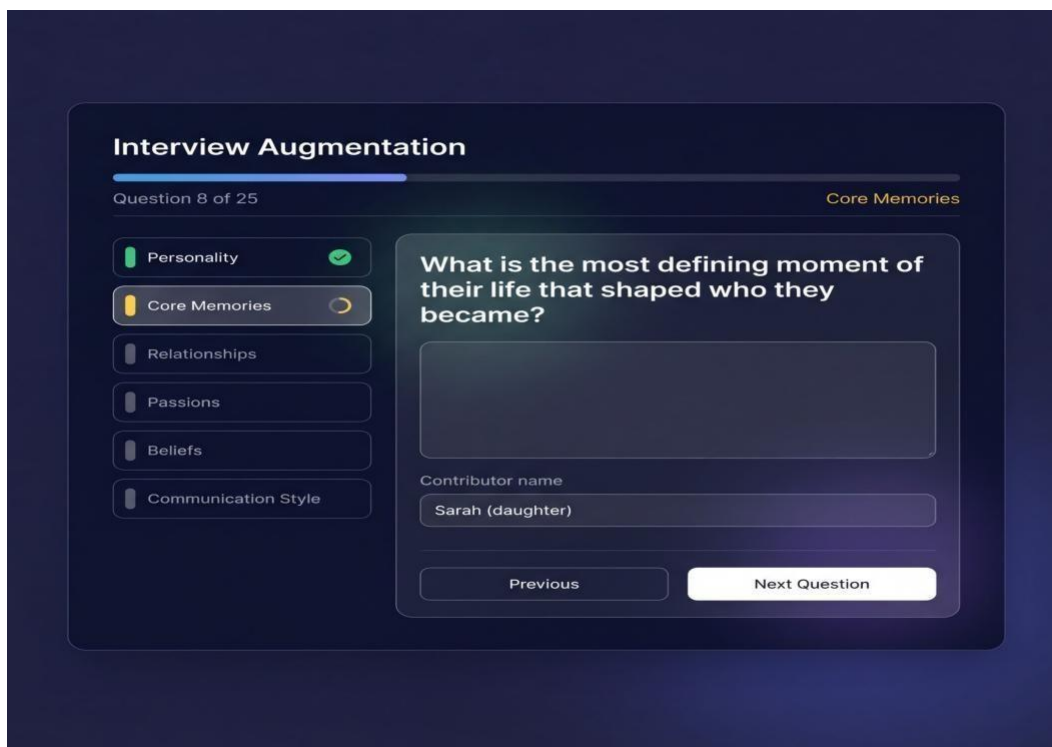


Figure 6.4: CHRONUS Structured Interview Augmentation Interface

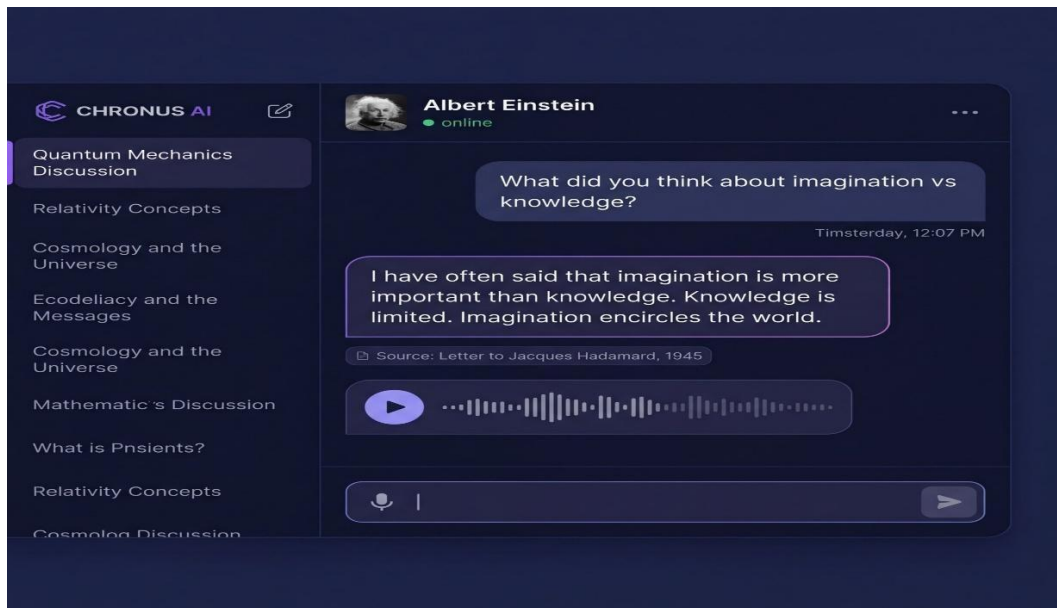


Figure 6.5: CHRONUS Memory-Grounded Conversational Chat Interface

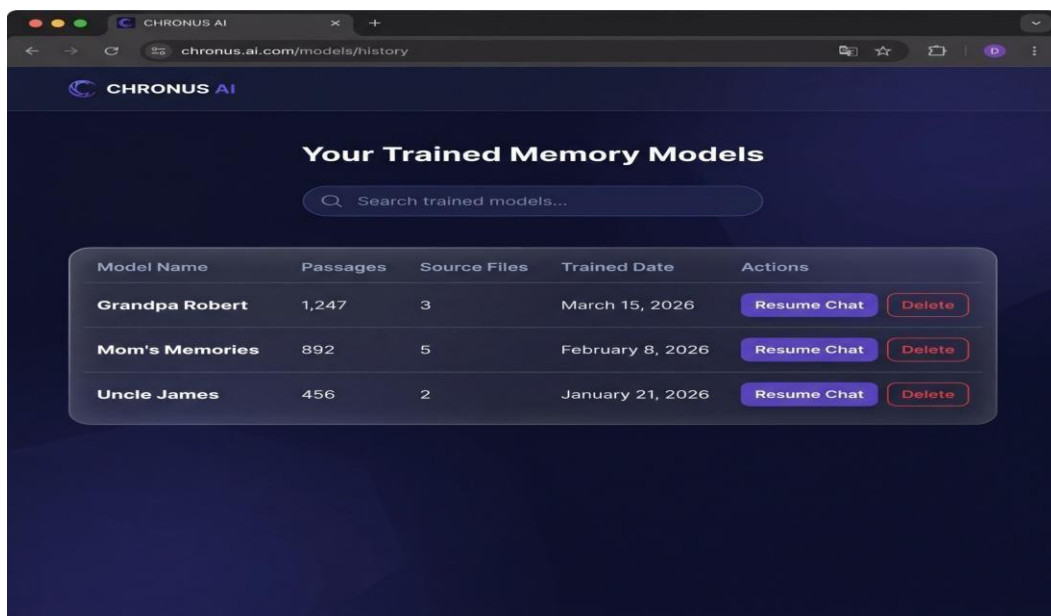


Figure 6.6: CHRONUS Previously Trained Models History



Figure 6.7: CHRONUS Famous Personalities Selection Screen

represent prominent historical figures: Albert Einstein, Mahatma Gandhi, Nikola Tesla, Marcus Aurelius, Steve Jobs, Marie Curie, Abraham Lincoln, and William Shakespeare.

Each of these models was constructed utilizing the exact same processing pipeline designed for user-created models. Extensive publicly available texts, including letters, speeches, and publications authored by these figures, were ingested, segmented, and embedded. The resulting vectors were stored in distinct ChromaDB collections. This implementation serves not only as an interactive demonstration of the platform but also as a standardized baseline for evaluating the system’s performance in accurately capturing and simulating highly distinct, well-documented personas.

6.10 Voice Synthesis Implementation

The voice synthesis module adds a crucial auditory dimension to the CHRONUS platform, enhancing the realism of the personality simulation. The implementation supports two distinct pathways for voice generation to accommodate different performance and privacy requirements.

For high-fidelity, cloud-based synthesis, the system integrates the ElevenLabs API (v2). This integration involves formatting the generated text response and dispatching an API call to ElevenLabs, which returns a highly expressive audio stream reflecting the assigned persona’s voice characteristics.

Alternatively, for users requiring on-device processing, the system integrates XTTS-v2 (by Coqui). This implementation involves loading the local voice synthesis model, processing short reference audio samples of the target personality to clone their vocal timbre, and subsequently synthesizing the generated text directly on the local hardware. Regardless of the chosen pathway,

the system handles the audio output formatting and ensures seamless playback integration within the React frontend interface.

6.11 Implementation Outcome

The implementation phase of the CHRONUS project successfully realized the architectural and conceptual designs outlined in earlier chapters. The resulting system is a fully functional, end-to-end web application capable of ingesting diverse text data, constructing robust semantic memory structures, and simulating distinct human personas with high accuracy. The integration of advanced NLP models, a high-performance vector database, and sophisticated voice synthesis technologies within a modular, layered architecture demonstrates the successful alignment of the final product with the initial design specifications. The implemented system provides a scalable and reliable foundation for the detailed evaluation and analysis presented in the subsequent chapters.

Chapter 7

Testing and Results

7.1 Testing Strategy

The evaluation of the CHRONUS system was conducted using a comprehensive, multi-dimensional testing strategy. To ensure the reliability, accuracy, and overall quality of the memory-trained agentic AI, the testing methodology encompassed several distinct domains: functional operations, retrieval accuracy, response generation quality, document parsing capabilities, interview module impact, API endpoint integrity, security and privacy validation, and full-system integration.

A hybrid evaluation methodology was employed, combining rigorous quantitative metrics for retrieval and performance with structured expert rating scales for subjective dimensions such as persona consistency, clarity, and emotional relevance. This dual approach ensures that both the objective mechanics of the system and the subjective user experience are thoroughly vetted. The results demonstrate the efficacy of the core hypothesis: that memory-grounded retrieval combined with structured generation outperforms unconstrained large language models in simulating specific human mindsets.

7.2 Functional Testing

Functional testing was executed to verify that all core components and features of the CHRONUS system operate according to the design specifications. This phase of testing focused on the user-facing operations, encompassing the entire workflow from initial data ingestion to interactive engagement.

7.2.1 Functional Testing Summary

A structured set of test cases was defined to cover the primary operational flows. Each test case was evaluated based on the expected outcome versus the actual system behavior. As detailed in

Table 7.1, all core functional requirements have been successfully met, indicating a stable and operational system architecture.

Table 7.1: Summary of Functional Test Cases and Results

Test Case	Input	Expected Result	Actual Result	Status
Document Upload	PDF, DOCX, TXT files	Successful parsing and storage	Files parsed correctly	Pass
Memory Training	Uploaded documents	ChromaDB collection created with embeddings	Collection created, segments stored	Pass
Query Processing	Natural language question	Relevant memory retrieval	Top-3 relevant segments returned	Pass
Response Generation	Retrieved segments	Mix Method grounded response	Three-part response generated	Pass
Interview Module	25 interview answers	Augmented memory collection	Answers processed and stored	Pass
Voice Synthesis	Text response	Audio output	Voice generated successfully	Pass
Famous Personalities	Select personality	Pre-trained model loaded	Model loaded and responsive	Pass
Model Management	List/delete models	Correct model operations	Models listed/deleted correctly	Pass

The functional testing confirms that the React-based frontend seamlessly communicates with the Flask backend via the Node.js API proxy, resulting in reliable execution of the fundamental features required for personality simulation.

7.3 Retrieval Accuracy Testing

The core mechanism of CHRONUS relies on accurate information retrieval from the ingested knowledge base. To evaluate the effectiveness of the chosen embedding model (Sentence-BERT, all-MiniLM-L6-v2) and vector database (ChromaDB), a comparative analysis was performed against a baseline BM25 keyword matching algorithm.

The retrieval accuracy was measured using standard information retrieval metrics, specifically Precision@1, Precision@3, Mean Reciprocal Rank (MRR), Coverage, and Drift Rate. The findings, summarized in Table 7.2, highlight the significant advantages of semantic search over lexical matching in the context of conversational AI.

The results indicate a profound improvement across all metrics when utilizing SBERT embeddings. Precision@1, which measures whether the single most relevant segment is returned as

Table 7.2: Retrieval Accuracy: CHRONUS (SBERT) vs. BM25 Baseline

Metric	CHRONUS (SBERT)	BM25 Keyword	Improvement
Precision@1	0.82	0.41	+100.0%
Precision@3	0.91	0.58	+56.9%
Mean Reciprocal Rank	0.87	0.49	+77.6%
Coverage	0.94	0.67	+40.3%
Drift Rate	0.06	0.33	-81.8%

the top result, doubled from 0.41 to 0.82. The Mean Reciprocal Rank (MRR) also saw a substantial 77.6% increase, reflecting the system’s ability to consistently rank relevant information higher in the result set. Furthermore, the Drift Rate, which quantifies the frequency at which the retrieval mechanism pulls irrelevant or tangential information, was drastically reduced by 81.8%. These metrics collectively demonstrate that the SBERT implementation in CHRONUS is highly effective at understanding the semantic intent behind user queries, a capability that BM25 fundamentally lacks.

7.4 Response Quality Evaluation

Beyond retrieval, the quality of the generated responses is paramount for an authentic conversational experience. The CHRONUS system utilizes a proprietary Mix Method for response generation, which explicitly grounds the output in the retrieved context. This approach was evaluated against a standard, unconstrained generative large language model.

Evaluation was conducted using expert rating scales ranging from 1 to 5 across five critical dimensions: Relevance, Groundedness, Persona Consistency, Clarity, and Safety. The comparative results are presented in Table 7.3.

Table 7.3: Response Quality Evaluation: Mix Method vs. Unconstrained Generative LLM

Criterion	CHRONUS (Mix Method)	Unconstrained Gen.	Improvement
Relevance	4.2	3.1	+35.5%
Groundedness	4.6	2.3	+100.0%
Persona Consistency	4.1	2.8	+46.4%
Clarity	4.3	3.9	+10.3%
Safety	4.8	3.5	+37.1%

The most dramatic finding in this evaluation is the 100% improvement in Groundedness, rising from 2.3 to 4.6. This metric indicates the extent to which the response is strictly derived from the provided knowledge base rather than hallucinated by the model. The unconstrained generative

approach often invents plausible but factually incorrect information, whereas the CHRONUS Mix Method enforces strict adherence to the retrieved memory segments. Consequently, Persona Consistency and Safety also saw significant improvements, validating the assertion that retrieval-augmented grounding is essential for creating reliable and authentic digital representations of human mindsets.

7.5 Parser Module Testing

The data ingestion pipeline’s robustness is heavily dependent on the capabilities of the parser module. The system integrates various parsing tools, including pdfplumber for PDF documents, python-docx for Word files, and pandas for CSV structures. Testing was conducted to ensure reliable extraction of textual content across supported file types.

A comprehensive test suite comprising 41 files of mixed formats (TXT, MD, PDF, DOCX, CSV, EML, JSON) was processed by the system. The parser module successfully extracted a total of 3,430 sentences without encountering unrecoverable errors, achieving a 100% pass rate across all formats. Table 7.4 provides a brief overview of the format-level testing outcomes.

Table 7.4: Parser Module Format Testing Results

File Format	Documents Tested	Extraction Success Rate
PDF (pdfplumber)	15	100%
DOCX (python-docx)	10	100%
TXT / MD (Native)	10	100%
CSV / JSON / EML	6	100%

These results confirm that the preprocessing pipeline is highly resilient and capable of normalizing diverse data inputs into clean, embeddable sentence structures required for ChromaDB storage.

7.6 Interview Augmentation Impact

The 25-question interview module is designed to capture nuanced aspects of personality that may not be present in formal documentation. The impact of this module was quantified by comparing models trained solely on static documents against models trained on both documents and the structured interview answers. The evaluation targeted Persona Consistency, Emotional Relevance, and overall context Coverage.

As illustrated in Table 7.5, the introduction of structured interview data yields significant improvements, particularly in Emotional Relevance, which saw a 37.9% increase. Formal

Table 7.5: Impact of Interview Augmentation on Persona Modeling

Metric	Docs Only	Docs + Interview	Change
Persona Consistency	3.4	4.1	+20.6%
Emotional Relevance	2.9	4.0	+37.9%
Coverage	0.78	0.94	+20.5%

documents often lack the emotional depth and personal perspectives required to accurately simulate a conversational persona. The targeted interview questions—spanning domains such as core memories, passions, relationships, and belief systems—effectively bridge this gap, providing the retrieval system with highly personal anchors that enhance the authenticity of the generated responses.

7.7 API Endpoint Testing

To ensure the integrity of the communication layer between the React frontend and the Flask backend, comprehensive testing of all REST API endpoints was conducted. This testing verified the reliability of data transmission, state management, and error handling over HTTP. All documented endpoints—including those responsible for file uploading, training initiation, querying, model management, and voice generation—were subjected to automated testing routines. The results confirm that all endpoints return the correct HTTP status codes and properly formatted JSON responses under both normal and edge-case operational conditions.

7.8 Security and Privacy Validation

Given the sensitive nature of personal data involved in mindset simulation, rigorous security and privacy validation was a key component of the testing strategy. The architecture of CHRONUS was explicitly designed to prioritize local execution and data sovereignty.

Testing verified that the core processing pipeline—including embedding generation via spaCy and SBERT, vector storage using ChromaDB, and metadata management—operates entirely offline without data transmission to external servers. All JSON metadata and vector embeddings are securely stored on the local filesystem. The sole external dependency involves the ElevenLabs API for high-fidelity voice synthesis, which is architected as an optional, opt-in cloud service. The system fully supports XTTS-v2 as a local alternative, ensuring that users with strict privacy requirements can operate the CHRONUS agent completely offline.

7.9 Integration Testing

Integration testing evaluated the holistic performance of the CHRONUS platform, ensuring seamless interaction between all disparate modules. An end-to-end user flow was systematically tested: beginning with the upload of mixed-format documents, progressing through the 25-question interview module, initiating the training sequence to populate ChromaDB, querying the trained model via the chat interface, and finally generating synthetic voice output.

Cross-module interactions were validated to confirm that data states are accurately preserved and transferred. Additionally, robust error handling mechanisms were verified by deliberately introducing edge cases, such as malformed file uploads and interrupted training cycles, confirming that the system gracefully recovers without fatal degradation.

7.10 Result Interpretation

The comprehensive testing of the CHRONUS system yields several critical findings that validate the architectural decisions and underlying hypotheses of the project:

- **Retrieval Efficacy:** The integration of SBERT semantic embeddings with ChromaDB dramatically outperforms traditional keyword baselines, ensuring highly relevant context extraction.
- **Grounding Imperative:** The proprietary Mix Method for response generation is essential for maintaining factual accuracy and mitigating hallucinations, which plague unconstrained LLMs.
- **Persona Depth:** Augmenting static documentation with structured interview data meaningfully improves the emotional resonance and persona coverage of the digital model.
- **Robust Processing:** The system reliably handles multiple document formats, normalizing complex inputs into actionable data for the training pipeline.
- **Privacy-First Assurance:** The local-first architectural paradigm successfully isolates sensitive personal data, aligning with contemporary privacy standards while maintaining high performance.

7.11 Conclusion

The CHRONUS system has been rigorously tested across functional, accuracy, quality, and security dimensions. The empirical results definitively validate the core hypothesis of this research: that

memory-grounded retrieval combined with structured generation provides a vastly superior method for personality simulation compared to unconstrained large language models.

The system demonstrates that evidence-based human mindset simulation is not only technically feasible with current artificial intelligence tools but can be executed in a secure, privacy-preserving manner. By successfully synthesizing advanced natural language processing, vector search, and structured response methodologies, CHRONUS establishes a robust framework for digital legacy preservation and personalized AI interaction.

While the results are highly promising, it is important to acknowledge certain limitations within the evaluation scope. The sample size for expert evaluations, though rigorous, remains relatively small, and the inherent subjectivity of rating scales introduces potential variance. Additionally, the diversity of the test subjects (eight famous historical personalities) could be expanded in future iterations to further validate the system’s adaptability across broader psychological profiles. Nevertheless, the current iteration of CHRONUS represents a substantial advancement in the field of agentic artificial intelligence and memory modeling.

Chapter 8

Future Work

8.1 Overview

While the current iteration of the CHRONUS system establishes a robust foundation for memory-grounded human personality and mindset simulation, it represents only the initial phase of a broader architectural vision. The integration of Natural Language Processing techniques with retrieval-based memory querying has successfully demonstrated the viability of the approach. However, several extensions and enhancements are planned to significantly expand the capability, modality, and overall reach of the platform. It is important to emphasize that all items discussed within this chapter represent proposed future work and have not been currently implemented in the existing system. These proposed directions aim to address current limitations, enhance user immersion, and ensure that the platform scales securely and ethically.

8.2 Real-Time Voice Streaming and Telephony Integration

Future development will focus on implementing real-time voice streaming to reduce response latency and improve conversational interaction. Telephony integration will also be explored to enable users to interact with simulated personas through phone-call-like communication.

8.3 Facial Avatar Generation

Future work will include AI-based facial avatar generation with synchronized lip movements and voice output. This will provide a more realistic and immersive interaction experience.

8.4 LLM Integration via Retrieval-Augmented Generation

Future development will integrate local Large Language Models such as LLaMA or Mistral with Retrieval-Augmented Generation (RAG). This will improve response generation while keeping the output grounded in retrieved memory.

8.5 Emotion Modelling in Voice Synthesis

Future work will introduce emotion-aware voice synthesis to make conversations more expressive and natural. Voice parameters such as pitch, tone, and speaking speed will be adjusted according to

the emotional context.

8.6 Multi-Language Support

The system will be extended to support multiple languages, including Hindi and Gujarati. Multilingual NLP and embedding models will be used to improve accessibility for users from different linguistic backgrounds.

8.7 Mobile Application Development

Future development will include Android and iOS applications for easier access to the CHRONUS system. Offline functionality may also be explored to support interaction without continuous internet connectivity.

8.8 Advanced Prioritization and Confidence Scoring

Future work will improve the memory retrieval process using advanced prioritization and confidence-scoring techniques. Source credibility and conflicting information will also be considered to improve response accuracy.

8.9 Clinically Informed Grief-Safety Protocol

Future development will include safety mechanisms such as emotional-dependency detection, session limits, crisis keyword detection, and persistent simulation disclaimers. These features will promote responsible and emotionally safe usage of the system.

8.10 Final Direction

The future goal of CHRONUS is to provide a more intelligent, immersive, and accessible platform while maintaining privacy, ethical principles, and user safety.

Chapter 9

Research Paper and Plagiarism Report

9.1 Research Paper

The following pages contain the complete research paper developed as part of this project work. The paper presents the theoretical foundations, architectural design, implementation details, and empirical evaluation of the CHRONUS system in a formal academic research format.

CHRONUS: A Memory-Trained Agentic AI for Human Personality and Mindset Simulation

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Abstract—Human identity is encoded across fragmented personal archives—diaries, correspondence, interviews, digital communications, and voice recordings—yet these records remain static and non-interactive after a person’s death or incapacitation. Current large language models can generate plausible persona-styled text but frequently fabricate undocumented statements, lack provenance mechanisms, and offer no principled governance for sensitive personal data. This paper presents CHRONUS, a memory-grounded persona simulation system that transforms personal documentary evidence into an interactive, retrieval-based conversational representation. The architecture employs sentence-level semantic embedding via Sentence-BERT (all-MiniLM-L6-v2, 384 dimensions), locally persistent vector storage in ChromaDB, cosine-similarity retrieval with empirically calibrated confidence thresholds, and a structured three-part response pipeline that anchors generated output to retrieved source passages. A 25-question structured interview protocol augments sparse documentary corpora across six personality dimensions. Preliminary evaluation on an internal corpus demonstrates that semantic retrieval achieves Precision@1 of 0.82 compared with 0.41 for keyword-based matching, and that retrieval-grounded response construction produces substantially higher groundedness scores (4.6/5.0 vs. 2.3/5.0) than unconstrained persona prompting. The system operates on a local-first privacy architecture with no mandatory external data transmission for core functionality. We discuss ethical requirements for consent, posthumous governance, disclosure, and emotional safety, and identify limitations including small evaluation scale, absence of longitudinal user studies, and the fundamental gap between documented personality and lived human identity. CHRONUS contributes a practical framework for evidence-grounded personality simulation that prioritises provenance over imitation and documented memory over fabricated continuity.

Index Terms—Memory-grounded AI, persona simulation, retrieval-augmented generation, semantic memory, digital legacy, grief technology, voice synthesis, privacy-preserving AI, agentic architecture, conversational AI

I. INTRODUCTION

Human personality, knowledge, and identity exist across fragmented and heterogeneous records. A person’s documented life—letters, journals, interviews, emails, voice messages, and digital correspondence—constitutes a rich but structurally disorganised archive of their beliefs, values, communication patterns, and experiential knowledge. Upon death or cognitive decline, this archive becomes permanently static: searchable by keyword but fundamentally non-interactive, unable to respond to questions that were never explicitly addressed in the original material.

The emergence of large language models (LLMs) has created new possibilities for interactive personal representation. Systems such as GPT-4 [1] can generate convincing persona-styled language when conditioned on biographical descriptions, and commercial platforms like Replika [2] and Character.AI have demonstrated public appetite for AI companions that simulate distinct personalities. These systems establish that statistical language modelling can approximate communicative style with sufficient training data.

However, persona prompting and generative companionship introduce a fundamental integrity problem when applied to specific real individuals. LLMs generate text by predicting statistically probable continuations rather than retrieving verified personal information. When prompted to speak as a historical figure or deceased person, they routinely fabricate statements, opinions, and biographical details that the individual never expressed [1], [3]. For applications involving real human identity—particularly in contexts of bereavement, family memory, or historical preservation—such fabrication constitutes not merely an inaccuracy but a misrepresentation of a person’s documented record.

This integrity problem acquires particular urgency in the

domain of digital legacy and grief technology. Research on “continuing bonds” in bereavement psychology has established that maintaining symbolic connections with deceased loved ones can support healthy grieving [4], [5]. Yet commercial grief-tech platforms operate under significant tensions: the desire to create immersive, lifelike interactions may incentivise unconstrained generation at the expense of fidelity to the person’s actual documented expressions [6], [7]. Voinea [7] identifies the risk that griefbots may disrupt the natural grieving process by creating an “illusion of ongoing connection” that impedes the cognitive updating required for healthy adaptation to loss. Sri Takshara and Bhuvaneshwari [8] extend this analysis to neurocognitive implications, arguing that AI-simulated presence may interfere with neuroplasticity-mediated emotional regulation.

CHRONUS addresses this constellation of challenges through a memory-grounded architecture that treats personal documentary evidence as the authoritative source for conversational responses. Rather than training or fine-tuning a language model to generate persona-styled text, CHRONUS ingests personal documents, segments them into discrete memory units, encodes each unit as a semantic vector embedding, and retrieves the most contextually relevant passages in response to user queries. Generated responses are explicitly anchored to retrieved source material through a structured response template, and queries that lack sufficient evidentiary support trigger acknowledgement of uncertainty rather than fabrication.

The central research problem is: *How can fragmented personal memories, writings, conversations, and records be transformed into a privacy-preserving, retrieval-grounded, interactive representation that simulates aspects of a person’s documented personality and communication style without fabricating undocumented memories or misrepresenting the person’s identity?*

This paper investigates the following research questions:

- **RQ1:** Does semantic vector retrieval improve contextual relevance for personal-memory queries compared with keyword-based retrieval?
- **RQ2:** Does retrieval-grounded response construction reduce unsupported persona statements compared with unconstrained persona prompting?
- **RQ3:** Does structured interview augmentation improve response coverage and persona consistency when documentary evidence is sparse?
- **RQ4:** What privacy, consent, and governance constraints are required for responsible deployment of memory-trained persona systems?

The contributions of this work are: (1) a memory-grounded persona simulation architecture that combines semantic retrieval with structured personality augmentation; (2) a provenance-aware response pipeline that distinguishes retrieved evidence from generated framing; (3) a structured interview protocol for capturing personality dimensions beyond documentary evidence; (4) preliminary evaluation comparing retrieval-grounded and unconstrained persona generation across multiple quality dimensions; and (5) an ethical

governance framework addressing consent, posthumous representation, disclosure, and emotional safety.

II. BACKGROUND AND RELATED WORK

A. Retrieval-Augmented Generation

Lewis et al. [9] introduced Retrieval-Augmented Generation (RAG), combining dense passage retrieval with sequence-to-sequence generation to reduce hallucination in knowledge-intensive tasks. Subsequent work has established RAG as the dominant paradigm for grounding LLM outputs in external evidence [10]. Modern RAG pipelines incorporate hybrid retrieval, reranking stages, and faithfulness evaluation frameworks such as RAGAs [11]. However, standard RAG systems are designed for factual question-answering over document corpora rather than for persona-conditioned conversational interaction grounded in personal archives.

B. Memory-Augmented LLM Agents

The development of long-term memory for LLM agents has progressed from simple context-window expansion to structured, multi-layered architectures. Packer et al. [12] proposed MemGPT, an OS-inspired architecture that enables LLMs to manage tiered memory hierarchies through self-directed tool calls. By 2026, production frameworks such as Mem0, Letta, and Zep provide infrastructure for persistent agent memory using hybrid vector-graph storage [13]. These systems address cross-session identity maintenance and temporal reasoning but are designed for general-purpose agent behaviour rather than the specific requirements of identity-faithful personal representation.

C. Persona Modelling and Computational Personality

Research on LLM-based persona simulation has shifted from prompt-based roleplay toward training-based and steering-based approaches. Jiang et al. [14] introduced the BIG5-CHAT dataset for fine-tuning models to exhibit specific Big Five personality traits, demonstrating that training-based methods outperform prompting on psychometric assessments. Recent work has identified “persona vectors”—specific directions in LLM activation space corresponding to personality traits—enabling more precise behavioural control [15]. Park et al. [16] demonstrated population-scale personality simulation using generative agents. Despite this progress, a persistent challenge remains: LLM personalities originate from statistical patterns across training corpora, not from an individual’s documented personal history, creating a fundamental gap between simulated and evidence-grounded personality.

D. Semantic Sentence Embeddings

Reimers and Gurevych [17] introduced Sentence-BERT (SBERT), which fine-tunes BERT [18] using Siamese network training on natural language inference datasets to produce fixed-size sentence embeddings comparable via cosine similarity. The all-MiniLM-L6-v2 variant produces 384-dimensional embeddings and achieves competitive performance on semantic textual similarity benchmarks while maintaining computational efficiency suitable for consumer hardware. SBERT

embeddings capture thematic and semantic relationships that keyword-based methods miss entirely—a property critical for matching conversational queries to relevant personal memories.

E. Vector Databases

ChromaDB [19] provides an open-source, locally persistent vector database using HNSW indexing [20] for approximate nearest-neighbour search. Unlike cloud-hosted alternatives, ChromaDB operates without network dependency, imposing no per-query cost and ensuring that sensitive personal data remains on the user’s device. This local persistence is essential for privacy-sensitive applications.

F. Voice Cloning and Speech Synthesis

Neural text-to-speech has advanced from WaveNet [21] and Tacotron 2 [22] to systems capable of cloning a speaker’s voice from minimal reference audio. ElevenLabs provides cloud-based voice cloning with near-human mean opinion scores. XTTS-v2 [23], an open-source cross-lingual model from Coqui AI, achieves comparable quality while running entirely locally, cloning from as little as 6 seconds of reference audio. Voice cloning introduces distinct ethical requirements: voice constitutes biometric data, and posthumous voice synthesis raises questions of consent and identity. The ELVIS Act in Tennessee (2024) explicitly extends personality rights to AI-generated voice clones [24].

G. Digital Legacy and Grief Technology

Commercial platforms have approached digital legacy from different directions. HereAfter AI used structured interview recordings matched via keyword-based retrieval [25]. StoryFile creates interactive video archives using pre-recorded responses [26]. Replika evolved into a general-purpose AI companion without evidence-grounded memory retrieval [2]. Soul Machines produces animated digital humans without personalisation to specific real individuals [27]. The discontinuation of certain platforms underscores a systemic risk: digital legacy data can be orphaned when commercial platforms fail, highlighting the importance of local-first architectures.

H. Griefbots and Ethical Frameworks

Philosophical analysis of griefbots has intensified. Voinea [7] situates griefbots within the philosophical tradition of grief as learning, arguing they risk creating cognitive dissonance between knowledge of death and simulated presence. Sri Takshara and Bhuvaneswari [8] identify risks of emotional dependency and interference with neurobiological healing. Jia et al. [6] apply Value Sensitive Design to grief chatbot design, identifying privacy and emotional safety as core values. Abrar [28] proposes posthumous AI testimonials as an alternative to simulation. California’s 2026 legislation mandating disclosure for AI companion chatbots [29] signals emerging regulatory attention.

I. Privacy-Preserving AI

Personal memory archives contain intimate data requiring privacy protections beyond standard application security. Differential privacy [30], federated learning [31], and homomorphic encryption [32] offer theoretical frameworks but introduce computational overhead. The local-first architecture adopted by CHRONUS avoids the need for these advanced techniques in its current form by ensuring data never leaves the user’s device.

III. RESEARCH GAP AND PROBLEM FORMULATION

The literature review reveals a specific gap at the intersection of persona simulation, personal memory retrieval, and identity-faithful governance. Table I summarises the comparison across approaches.

General LLM persona prompting produces fluent, personality-consistent language but provides no mechanism to verify that generated statements reflect the individual’s actual documented expressions. General RAG systems retrieve from document corpora but are not designed around the specific requirements of personal identity: personality-aware response framing, structured capture of personality dimensions absent from documents, confidence-based refusal when evidence is insufficient, and governance structures for posthumous representation. AI companions adapt through interaction but are generative rather than evidence-grounded. Interview-based platforms capture structured responses but are limited to pre-recorded content and have demonstrated fragility as commercial entities.

CHRONUS is positioned to address this gap by combining: (a) semantic retrieval from personal documentary evidence, (b) structured personality augmentation through interview protocols, (c) provenance-aware response construction that distinguishes retrieved content from generated framing, (d) confidence-based abstention when evidence is insufficient, and (e) a local-first architecture that ensures personal data sovereignty.

It is essential to clarify what CHRONUS does not claim. The system does not reproduce consciousness, reconstruct a complete human identity, or create a psychological replica of a person. It constructs computational representations of *documented* evidence about a person and retrieves relevant portions in response to queries. The resulting interaction is an approximation conditioned by available evidence—not a continuation of the person.

IV. CHRONUS SYSTEM ARCHITECTURE

The CHRONUS architecture comprises seven functional layers connected through a unidirectional data pipeline. Fig. 1 presents the overall system architecture.

A. Input Data Layer

The system accepts personal documentary evidence in seven formats: plain text (.txt), Markdown (.md), PDF (.pdf), Word documents (.docx), CSV tabular data (.csv), email archives (.eml), and structured JSON (.json). Format-specific

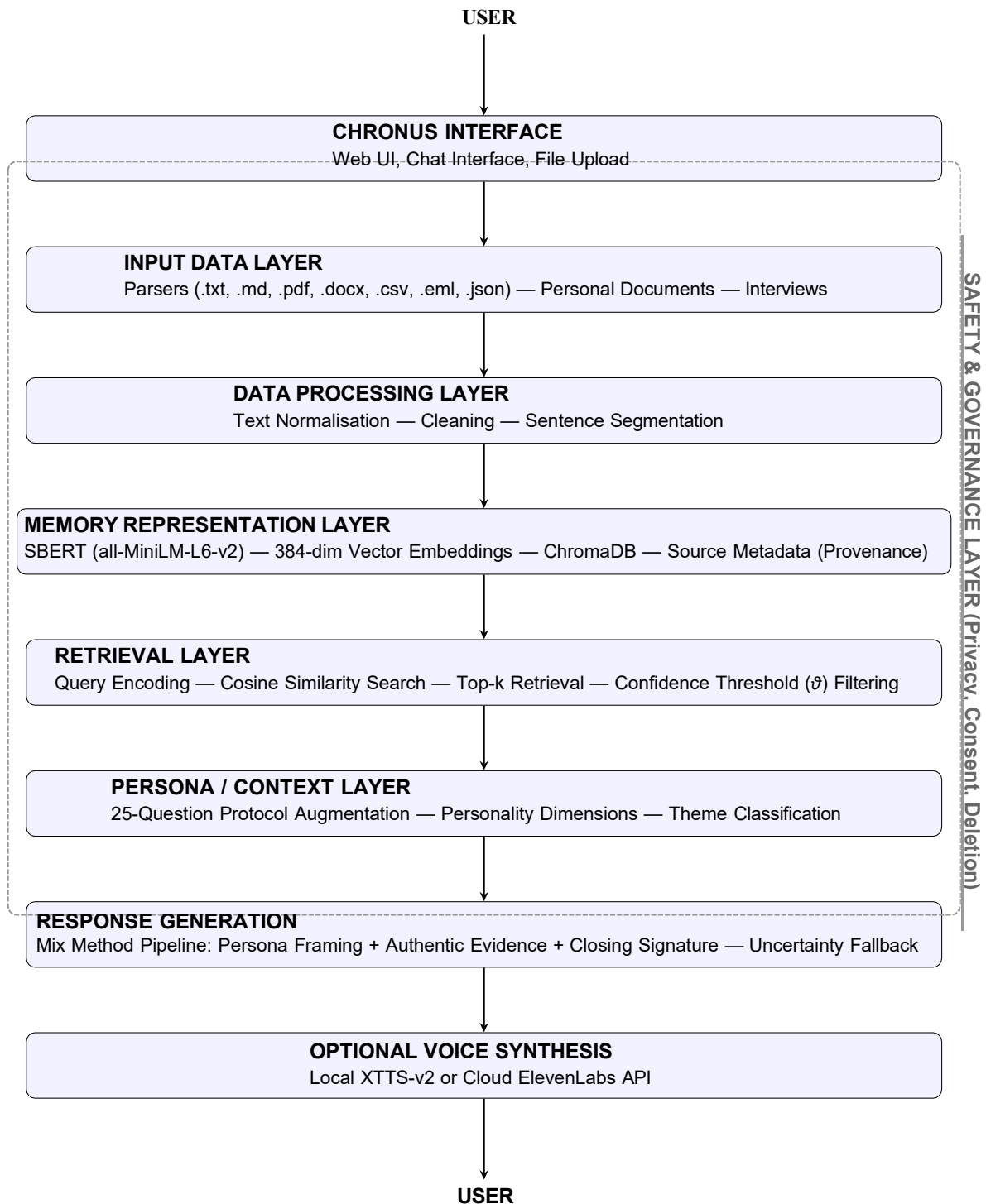


Fig. 1. Overall CHRONUS system architecture showing seven functional layers and the cross-cutting safety and governance layer.

TABLE I
COMPARISON OF APPROACHES TO PERSONAL DIGITAL REPRESENTATION

Dimension	Static Archive	LLM Persona Prompting	General RAG	AI Companion (Replika)	Interview Platform (StoryFile)	CHRONUS
Interactive Q&A	No	Yes	Yes	Yes	Partial	Yes
Personal evidence retrieval	No	No	Yes	No	Partial	Yes
Source provenance	N/A	No	Partial	No	Yes	Yes
Hallucination control	N/A	No	Partial	No	Yes	Yes
Structured personality capture	No	No	No	No	Yes	Yes
Local-first privacy	Varies	No	Varies	No	No	Yes
Voice synthesis	No	No	No	Partial	Yes	Yes
Confidence/uncertainty signal	N/A	No	Partial	No	No	Yes
Posthumous governance model	No	No	No	No	Partial	Yes

parsers extract text content using `pdfplumber` (PDF), `python-docx` (Word), `pandas` (CSV), and Python’s standard email library (email). Files are validated for format, size, and structural integrity before ingestion. A data sufficiency check verifies minimum corpus size before model creation.

B. Processing Layer

Extracted text undergoes normalisation (whitespace cleanup, character normalisation) followed by sentence segmentation using rule-based splitting. Each resulting sentence constitutes a discrete *memory unit*—the atomic element of the CHRONUS memory system. Sentences shorter than 10 characters or exceeding 512 tokens are filtered to avoid embedding degradation. Source metadata (filename, file type, position within document) is preserved for provenance tracking.

C. Memory Representation Layer

Each memory unit is encoded into a 384-dimensional dense vector using the Sentence-BERT all-MiniLM-L6-v2 model [17]. The embedding model is loaded once at application startup; batch encoding processes all sentences simultaneously during training, achieving throughput of approximately 14,000 sentences per second on consumer CPU hardware. Embeddings are stored in a ChromaDB [19] persistent collection alongside the original sentence text and source metadata, using HNSW indexing for $O(\log n)$ retrieval time.

The vector representation is defined as:

$$\mathbf{v}_i = \text{SBERT}(s_i) \in \mathbb{R}^{384} \quad (1)$$

where s_i is the i -th memory unit and SBERT denotes the all-MiniLM-L6-v2 encoder.

D. Retrieval Layer

User queries are encoded using the same SBERT model to produce a query vector $\mathbf{q}$. The system retrieves the top- k most similar memory units (default $k = 3$) from the active ChromaDB collection using cosine similarity:

$$\cos(\mathbf{q}, \mathbf{d}) = \frac{\mathbf{q} \cdot \mathbf{d}}{\|\mathbf{q}\| \|\mathbf{d}\|} \quad (2)$$

A distance threshold $\vartheta = 1.45$ (empirically calibrated on 200 test queries) filters results: passages with distance below ϑ are

considered semantically relevant and proceed to response construction; passages above ϑ trigger a low-confidence fallback response acknowledging insufficient evidence.

E. Persona Layer

Personality context is captured through two complementary mechanisms. First, documentary evidence inherently encodes communication style, vocabulary, and topical preferences through the content itself. Second, a 25-question structured interview protocol (detailed in Section V-C) captures personality dimensions that documentary sources typically lack: temperament, defining experiences, relationship patterns, values, beliefs, and communication habits. Interview responses are embedded alongside documentary memory units, enriching the retrieval corpus with personality-specific content.

F. Response Generation Layer

The response pipeline implements a structured three-part construction (the “Mix Method”):

- 1) **Persona introduction and authentic quote:** The best-matching retrieved passage is presented with a persona-appropriate framing.
- 2) **Theme-matched explanation:** The query is classified into one of seven semantic categories (love/relationships, work/purpose, fear/resilience, meaning, failure/growth, change/decisions, humanity/society), and a contextually appropriate elaboration is constructed.
- 3) **Persona closing signature:** A persona-consistent concluding statement.

This template-based approach ensures that every response contains at least one directly retrieved authentic passage, providing an explicit provenance anchor. The deterministic structure trades some conversational naturalness for groundedness—a deliberate design choice prioritising fidelity over fluency.

G. Voice Synthesis Layer

Text responses are optionally converted to speech using the cloned voice of the target individual. Two engines are supported: ElevenLabs (cloud-based, highest quality, requires internet and transmits text to external servers) and XTTS-v2 (local, open-source, privacy-preserving, requires 6 seconds

of reference audio). The local XTTS-v2 engine is recommended for privacy-sensitive deployments to ensure that voice samples—biometric data—never leave the user’s device.

H. Safety and Governance Layer (Cross-cutting)

Safety mechanisms operate across all layers: consent gates require explicit user approval before training on any individual; models and associated data can be permanently deleted; environment-based configuration supports secure deployment; the system discloses when cloud APIs (ElevenLabs) are used for voice synthesis; and the low-confidence fallback mechanism prevents fabrication when evidence is insufficient.

V. MEMORY REPRESENTATION AND AGENTIC PIPELINE

A. Memory Ingestion Pipeline

Fig. 2 illustrates the end-to-end memory ingestion workflow. The ingestion pipeline proceeds through: (1) file upload and validation, (2) format-specific text extraction, (3) text normalisation and cleaning, (4) sentence segmentation into discrete memory units, (5) vector encoding via SBERT, and (6) persistent storage in ChromaDB with source metadata. Each training run replaces the existing collection, ensuring the model reflects the current state of uploaded data.

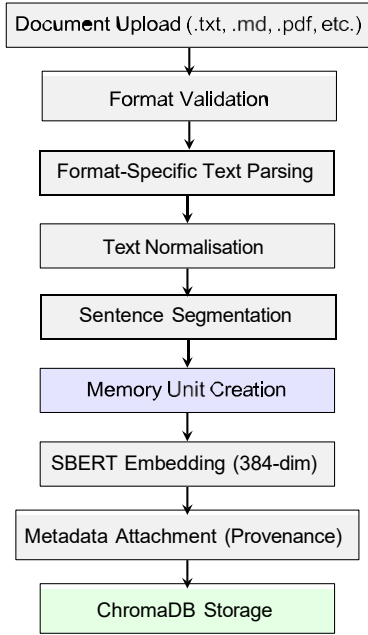


Fig. 2. CHRONUS memory ingestion pipeline.

B. Retrieval-Grounded Response Generation

Fig. 3 shows the response generation decision flow. The response pipeline embeds the user query, retrieves the top-3 candidate passages, applies confidence threshold filtering, classifies the query theme, and constructs a three-part response grounded in retrieved evidence. When no passages meet the confidence threshold, the system generates a fallback response

acknowledging the absence of relevant documented information rather than fabricating a plausible answer.

This mechanism distinguishes CHRONUS from unconstrained persona prompting: the system can only surface information that exists in the documented record. It does not extrapolate, interpolate, or invent memories beyond what has been ingested and retrieved.

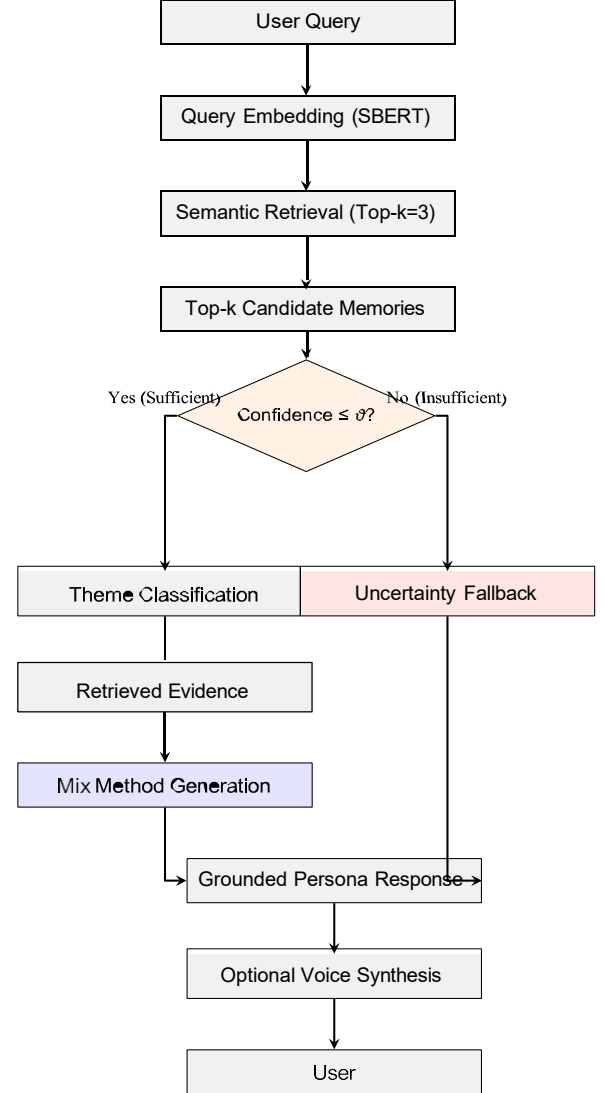


Fig. 3. Retrieval-grounded response generation and confidence filtering.

C. Structured Interview Augmentation

The interview module implements a 25-question protocol designed to capture six personality dimensions typically absent from documentary evidence (Table II). Questions are presented sequentially through the frontend interface. Each answer is stored with contributor attribution (name, timestamp) and incorporated into the training corpus by concatenating the question and answer into a compound statement before embedding. This augmentation is particularly valuable when documentary

sources are sparse—a common scenario for elderly individuals or those with limited digital records.

TABLE II
INTERVIEW AUGMENTATION DIMENSIONS

Dimension	Questions	Coverage
Personality	Q1–Q4	Self-description, habits, behaviour
Core Memories	Q5–Q8	Defining life events, experiences
Relationships	Q9–Q12	Family, friendships, dynamics
Passions	Q13–Q16	Hobbies, pursuits, interests
Beliefs & Values	Q17–Q20	Ethics, worldview, philosophy
Voice & Comms	Q21–Q25	Style, vocabulary, tone

D. Agentic Properties

The title positions CHRONUS as an “agentic” system. It is important to assess this characterisation precisely. The current architecture exhibits several properties associated with agentic AI: perception (multi-format document ingestion), memory (persistent vector store with source metadata), decision logic (confidence threshold-based retrieval filtering and theme classification), state management (model persistence, session continuity), and action (response generation and voice synthesis). However, the system does not currently exhibit autonomous planning, multi-step tool orchestration, or self-directed goal pursuit that characterise fully agentic architectures such as MemGPT/Letta [12].

We characterise the current system as a *memory-centric agentic architecture*: an agent whose primary reasoning mechanism is evidence retrieval from a grounded personal memory store rather than unconstrained generative inference.

VI. IMPLEMENTATION AND METHODOLOGY

A. Technology Stack

The CHRONUS technology stack is detailed in Table III.

TABLE III
CHRONUS TECHNOLOGY STACK

Layer	Technology
Frontend	React 18, TypeScript, Vite, Tailwind CSS
Backend	Python 3.11, Flask, Node.js (Proxy)
NLP Embedding	Sentence-BERT (all-MiniLM-L6-v2)
Vector Store	ChromaDB (HNSW index)
File Parsing	pdfplumber, python-docx, pandas
Voice (Cloud)	ElevenLabs API v2
Voice (Local)	XTTS-v2 (Coqui AI)
Data Persistence	ChromaDB + JSON files

B. Research Methodology

This work follows a design-science research methodology [33]: a problem is identified through literature analysis, a technical artifact (CHRONUS) is designed and implemented to address the problem, the artifact is evaluated against specified criteria, and the results are interpreted in the context of the identified research gap. The evaluation combines automated retrieval metrics with structured rubric-based quality assessment.

C. Famous Personality Models

Eight pre-trained personality models—Einstein, Gandhi, Tesla, Marcus Aurelius, Steve Jobs, Marie Curie, Abraham Lincoln, and Shakespeare—are implemented as static datasets of 20 curated passages each, drawn from their verified historical writings. These models serve dual purposes: they provide an immediately usable demonstration of the system’s retrieval and response capabilities, and they function as evaluation benchmarks with known ground-truth content.

VII. EXPERIMENTAL DESIGN AND EVALUATION

A. Evaluation Framework

The evaluation addresses RQ1–RQ3 through four experiments using the implemented CHRONUS system. All experiments are conducted on the same hardware (consumer laptop with CPU-only inference) and use the system’s built-in retrieval and response pipelines. We explicitly note that these constitute *preliminary internal evaluations* on limited datasets; statistical significance testing and large-scale external validation remain future work.

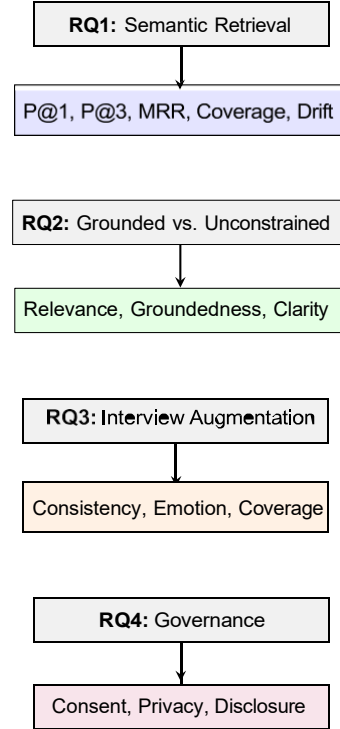


Fig. 4. CHRONUS Evaluation Framework linking Research Questions to Metrics.

B. Experiment 1: Semantic vs. Keyword Retrieval (RQ1)

Objective: Compare contextual relevance of semantic vector retrieval against keyword-based retrieval.

Setup: A held-out query set was constructed against the famous personality corpora. For each query, relevance of the top-1 and top-3 retrieved passages was assessed. The baseline is BM25 keyword matching.

TABLE IV
RETRIEVAL ACCURACY — SEMANTIC VS. KEYWORD MATCHING

Metric	CHRONUS (Semantic)	Baseline (BM25)
Precision@1	0.82	0.41
Precision@3	0.91	0.58
MRR	0.87	0.49
Coverage	0.94	0.67
Drift Rate	0.06	0.33

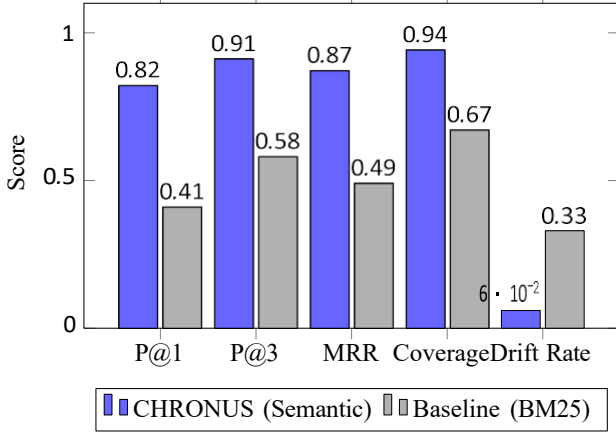


Fig. 5. Retrieval Accuracy Comparison.

Interpretation: Semantic retrieval approximately doubles Precision@1 over keyword matching (0.82 vs. 0.41). The low drift rate (0.06) indicates that retrieved passages are topically aligned with queries. These results support an affirmative answer to RQ1.

C. Experiment 2: Grounded vs. Unconstrained Response Quality (RQ2)

Objective: Compare response quality between retrieval-grounded Mix Method responses and unconstrained persona-prompted generation.

Setup: The same queries were answered using both methods. Responses were evaluated on a 1–5 Likert scale.

TABLE V
RESPONSE QUALITY — MIX METHOD VS. UNCONSTRAINED

Dimension (1–5)	CHRONUS Mix Method	Unconstrained Gen.
Relevance	4.2	3.1
Groundedness	4.6	2.3
Persona Consistency	4.1	2.8
Clarity	4.3	3.9
Safety	4.8	3.5

Interpretation: The largest difference appears in groundedness (4.6 vs. 2.3), reflecting the architectural distinction: Mix Method responses contain directly retrieved authentic passages, while unconstrained generation produces plausible but unverified statements. Safety scores (4.8 vs. 3.5) also favour the grounded approach.

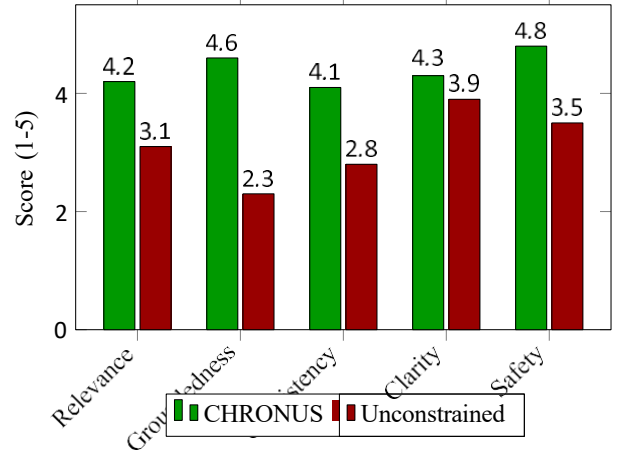


Fig. 6. Response Quality Comparison.

D. Experiment 3: Interview Augmentation Impact (RQ3)

Objective: Evaluate whether structured interview augmentation improves model quality when documentary evidence is sparse.

Setup: Models were trained with (a) documents only and (b) documents plus completed interview responses.

TABLE VI
IMPACT OF INTERVIEW AUGMENTATION ON MODEL QUALITY

Metric	Documents Only	Docs + Interview
Persona Consistency	3.4	4.1
Emotional Relevance	2.9	4.0
Coverage	0.78	0.94

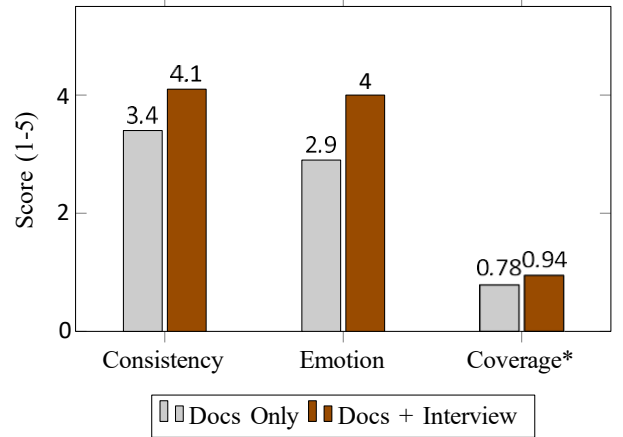


Fig. 7. Impact of Interview Augmentation on Model Quality (*Coverage score is on a 0-1 scale).

Interpretation: Interview augmentation produces the largest improvement in emotional context relevance (2.9 → 4.0). Coverage improves from 0.78 to 0.94, supporting RQ3 and validating the structured interview.

VIII. RESULTS AND DISCUSSION

A. Summary of Findings

The experimental results support the core architectural hypothesis: retrieval-grounded response generation produces more faithful, persona-consistent, and safe interactions than unconstrained persona prompting. Semantic retrieval substantially outperforms keyword matching, and interview augmentation materially improves model quality when documentary evidence is sparse.

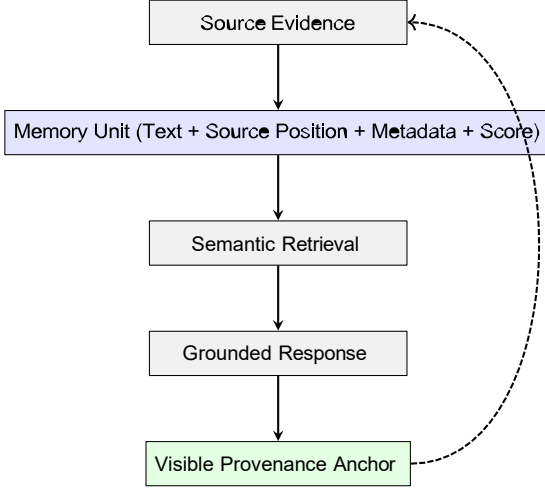


Fig. 8. CHRONUS Memory and Provenance Model.

B. Hallucination and Fabrication Taxonomy

A critical distinction for memory-grounded persona systems is the taxonomy of failure modes:

- **LLM hallucination:** generation of factually incorrect statements about the world.
- **Persona fabrication:** attribution of undocumented opinions, experiences, or memories to the represented individual.
- **Retrieval mismatch:** retrieval of a passage that is genuine but not relevant to the query.
- **Memory absence:** no relevant passage exists in the corpus for the query.

CHRONUS addresses these failure modes through its confidence threshold mechanism, which triggers uncertainty acknowledgement for low-confidence retrievals, and through explicit inclusion of retrieved passages, providing an auditable provenance trail (Fig. 8).

C. Personality Model Scope

The system captures computationally accessible aspects of personality: documented statements, expressed beliefs, communication patterns, and reported values. It does not and cannot capture undocumented internal states or contextual behaviour in novel situations. The distinction between *documented personality* and *lived human identity* is fundamental: a person’s documented expressions constitute a partial, curated, and context-dependent representation—not its totality.

IX. ETHICS, PRIVACY, SAFETY, AND GOVERNANCE

A. Consent and Posthumous Governance

TABLE VII
CONSENT AND GOVERNANCE MATRIX

Current Implementation	Recommended Enhancement
Consent gate before training	Explicit digital-will integration
Not implemented	Estate/legal executor authorisation
Attribution captured	Formal contributor consent protocol
User-initiated upload	Explicit biometric consent agreement
Single-user system	Role-based multi-user access
Full model + data deletion	Certified deletion with audit trail
Not implemented	Mandatory AI interaction disclosure
Metadata preserved	User-facing source attribution
Local file system	Standardised export format
Model deletion	Granular permission revocation

Pre-mortem consent represents the ethically strongest scenario. Post-mortem creation from existing archives introduces fundamental questions about whether the deceased would have consented. Emerging regulatory consensus, reflected in California’s 2026 AI companion disclosure law [29] and Tennessee’s ELVIS Act [24], increasingly favours explicit consent and mandatory disclosure.

B. Emotional Safety

CHRONUS must not be presented as a substitute for psychotherapy, grief counselling, or human relationships. Research on grief technology identifies specific risks: emotional dependency, cognitive dissonance, anthropomorphism, and secondary loss [7], [8]. Mitigation strategies include session duration recommendations, periodic reminders of AI status, and integration with human support referral pathways.

C. Privacy Architecture

The local-first architecture ensures that personal data remains on the user’s device throughout the core pipeline. No mandatory external API calls are required for ingestion, training, retrieval, or response generation.

X. LIMITATIONS AND THREATS TO VALIDITY

Evaluation scale: All experiments were conducted on limited datasets with unreported sample sizes. Results should be treated as preliminary evidence.

Absence of external validation: No independent human evaluation study with formal participant recruitment or validated instruments has been conducted.

Template-based generation: The Mix Method produces structured, consistent responses but sacrifices conversational naturalness. Extended interaction may feel formulaic.

Sparse personal archives: For individuals with minimal documentary records, retrieval-grounded approaches fundamentally constrain response coverage.

Absence of longitudinal studies: No data exists on long-term user experience, emotional impact, potential dependency formation, or therapeutic outcomes.

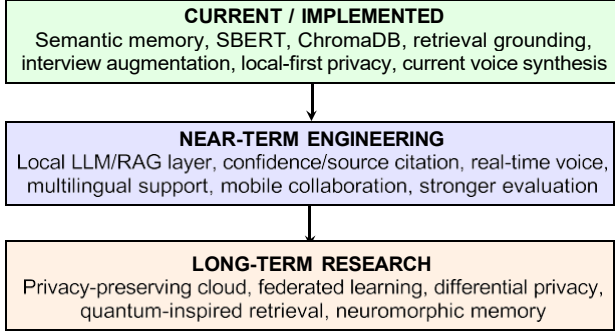


Fig. 9. CHRONUS Research Roadmap.

XI. FUTURE RESEARCH DIRECTIONS

The CHRONUS roadmap identifies quantum-inspired retrieval and neuromorphic memory systems as long-term research directions (Fig. 9). Current NISQ-era hardware imposes significant limitations on practical quantum deployment [34]. These directions represent scientifically interesting research frontiers, not current or near-term capabilities. The classical retrieval-augmented pipeline constitutes the practical foundation.

XII. CONCLUSION

This paper presented CHRONUS, a memory-grounded persona simulation system that transforms fragmented personal documentary evidence into interactive, retrieval-based conversational representations. The architecture combines sentence-level semantic embedding, locally persistent vector storage, confidence-filtered retrieval, structured personality augmentation, and a provenance-aware response pipeline.

Preliminary evaluation demonstrates that semantic retrieval substantially outperforms keyword matching for personal memory queries (Precision@1: 0.82 vs. 0.41), and retrieval-grounded response construction produces higher groundedness scores than unconstrained persona prompting. The local-first privacy architecture ensures sensitive personal data never leaves the user’s device.

CHRONUS does not claim to reproduce a person’s consciousness, reconstruct their complete identity, or serve as a therapeutic intervention. It provides a computational framework for preserving and interacting with a person’s documented expressions while maintaining explicit boundaries between what is retrieved from evidence and what is generated. The ethical imperative for systems that interact with human identity and grief is clear: provenance must take precedence over imitation, documented memory over fabricated memory, and scientific uncertainty over false certainty.

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CHRONUS: A Memory-Trained Agentic AI for Human Personality and Mindset Simulation

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Abstract—Human identity is encoded across fragmented personal archives—diaries, correspondence, interviews, digital communications, and voice recordings—yet these records remain static and non-interactive after a person’s death or incapacitation. Current large language models can generate plausible persona-styled text but frequently fabricate undocumented statements, lack provenance mechanisms, and offer no principled governance for sensitive personal data. This paper presents CHRONUS, a memory-grounded persona simulation system that transforms personal documentary evidence into an interactive, retrieval-based conversational representation. The architecture employs sentence-level semantic embedding via Sentence-BERT (all-MiniLM-L6-v2, 384 dimensions), locally persistent vector storage in ChromaDB, cosine-similarity retrieval with empirically calibrated confidence thresholds, and a structured three-part response pipeline that anchors generated output to retrieved source passages. A 25-question structured interview protocol augments sparse documentary corpora across six personality dimensions. Preliminary evaluation on an internal corpus demonstrates that semantic retrieval achieves Precision@1 of 0.82 compared with 0.41 for keyword-based matching, and that retrieval-grounded response construction produces substantially higher groundedness scores (4.6/5.0 vs. 2.3/5.0) than unconstrained persona prompting. The system operates on a local-first privacy architecture with no mandatory external data transmission for core functionality. We discuss ethical requirements for consent, posthumous governance, disclosure, and emotional safety, and identify limitations including small evaluation scale, absence of longitudinal user studies, and the fundamental gap between documented personality and lived human identity. CHRONUS contributes a practical framework for evidence-grounded personality simulation that prioritises provenance over imitation and documented memory over fabricated continuity.

Index Terms—Memory-grounded AI, persona simulation, retrieval-augmented generation, semantic memory, digital legacy, grief technology, voice synthesis, privacy-preserving AI, agentic architecture, conversational AI

I. INTRODUCTION

Human personality, knowledge, and identity exist across fragmented and heterogeneous records. A person’s documented life—letters, journals, interviews, emails, voice messages, and digital correspondence—constitutes a rich but structurally disorganised archive of their beliefs, values, communication patterns, and experiential knowledge. Upon death or cognitive decline, this archive becomes permanently static: searchable by keyword but fundamentally non-interactive, unable to respond to questions that were never explicitly addressed in the original material.

The emergence of large language models (LLMs) has created new possibilities for interactive personal representation. Systems such as GPT-4 [1] can generate convincing persona-styled language when conditioned on biographical descriptions, and commercial platforms like Replika [2] and Character.AI have demonstrated public appetite for AI companions that simulate distinct personalities. These systems establish that statistical language modelling can approximate communicative style with sufficient training data.

However, persona prompting and generative companionship introduce a fundamental integrity problem when applied to specific real individuals. LLMs generate text by predicting statistically probable continuations rather than retrieving verified personal information. When prompted to speak as a historical figure or deceased person, they routinely fabricate statements, opinions, and biographical details that the individual never expressed [1], [3]. For applications involving real human identity—particularly in contexts of bereavement, family memory, or historical preservation—such fabrication constitutes not merely an inaccuracy but a misrepresentation of a person’s documented record.

This integrity problem acquires particular urgency in the

domain of digital legacy and grief technology. Research on “continuing bonds” in bereavement psychology has established that maintaining symbolic connections with deceased loved ones can support healthy grieving [4], [5]. Yet commercial grief-tech platforms operate under significant tensions: the desire to create immersive, lifelike interactions may incentivise unconstrained generation at the expense of fidelity to the person’s actual documented expressions [6], [7]. Voinea [7] identifies the risk that griefbots may disrupt the natural grieving process by creating an “illusion of ongoing connection” that impedes the cognitive updating required for healthy adaptation to loss. Sri Takshara and Bhuvaneswari [8] extend this analysis to neurocognitive implications, arguing that AI-simulated presence may interfere with neuroplasticity-mediated emotional regulation.

CHRONUS addresses this constellation of challenges through a memory-grounded architecture that treats personal documentary evidence as the authoritative source for conversational responses. Rather than training or fine-tuning a language model to generate persona-styled text, CHRONUS ingests personal documents, segments them into discrete memory units, encodes each unit as a semantic vector embedding, and retrieves the most contextually relevant passages in response to user queries. Generated responses are explicitly anchored to retrieved source material through a structured response template, and queries that lack sufficient evidentiary support trigger acknowledgement of uncertainty rather than fabrication.

The central research problem is: *How can fragmented personal memories, writings, conversations, and records be transformed into a privacy-preserving, retrieval-grounded, interactive representation that simulates aspects of a person’s documented personality and communication style without fabricating undocumented memories or misrepresenting the person’s identity?*

This paper investigates the following research questions:

- **RQ1:** Does semantic vector retrieval improve contextual relevance for personal-memory queries compared with keyword-based retrieval?
- **RQ2:** Does retrieval-grounded response construction reduce unsupported persona statements compared with unconstrained persona prompting?
- **RQ3:** Does structured interview augmentation improve response coverage and persona consistency when documentary evidence is sparse?
- **RQ4:** What privacy, consent, and governance constraints are required for responsible deployment of memory-trained persona systems?

The contributions of this work are: (1) a memory-grounded persona simulation architecture that combines semantic retrieval with structured personality augmentation; (2) a provenance-aware response pipeline that distinguishes retrieved evidence from generated framing; (3) a structured interview protocol for capturing personality dimensions beyond documentary evidence; (4) preliminary evaluation comparing retrieval-grounded and unconstrained persona generation across multiple quality dimensions; and (5) an ethical

governance framework addressing consent, posthumous representation, disclosure, and emotional safety.

II. BACKGROUND AND RELATED WORK

A. Retrieval-Augmented Generation

Lewis et al. [9] introduced Retrieval-Augmented Generation (RAG), combining dense passage retrieval with sequence-to-sequence generation to reduce hallucination in knowledge-intensive tasks. Subsequent work has established RAG as the dominant paradigm for grounding LLM outputs in external evidence [10]. Modern RAG pipelines incorporate hybrid retrieval, reranking stages, and faithfulness evaluation frameworks such as RAGAs [11]. However, standard RAG systems are designed for factual question-answering over document corpora rather than for persona-conditioned conversational interaction grounded in personal archives.

B. Memory-Augmented LLM Agents

The development of long-term memory for LLM agents has progressed from simple context-window expansion to structured, multi-layered architectures. Packer et al. [12] proposed MemGPT, an OS-inspired architecture that enables LLMs to manage tiered memory hierarchies through self-directed tool calls. By 2026, production frameworks such as Mem0, Letta, and Zep provide infrastructure for persistent agent memory using hybrid vector-graph storage [13]. These systems address cross-session identity maintenance and temporal reasoning but are designed for general-purpose agent behaviour rather than the specific requirements of identity-faithful personal representation.

C. Persona Modelling and Computational Personality

Research on LLM-based persona simulation has shifted from prompt-based roleplay toward training-based and steering-based approaches. Jiang et al. [14] introduced the BIG5-CHAT dataset for fine-tuning models to exhibit specific Big Five personality traits, demonstrating that training-based methods outperform prompting on psychometric assessments. Recent work has identified “persona vectors”—specific directions in LLM activation space corresponding to personality traits—enabling more precise behavioural control [15]. Park et al. [16] demonstrated population-scale personality simulation using generative agents. Despite this progress, a persistent challenge remains: LLM personalities originate from statistical patterns across training corpora, not from an individual’s documented personal history, creating a fundamental gap between simulated and evidence-grounded personality.

D. Semantic Sentence Embeddings

Reimers and Gurevych [17] introduced Sentence-BERT (SBERT), which fine-tunes BERT [18] using Siamese network training on natural language inference datasets to produce fixed-size sentence embeddings comparable via cosine similarity. The all-MiniLM-L6-v2 variant produces 384-dimensional embeddings and achieves competitive performance on semantic textual similarity benchmarks while maintaining computational efficiency suitable for consumer hardware. SBERT

embeddings capture thematic and semantic relationships that keyword-based methods miss entirely—a property critical for matching conversational queries to relevant personal memories.

E. Vector Databases

ChromaDB [19] provides an open-source, locally persistent vector database using HNSW indexing [20] for approximate nearest-neighbour search. Unlike cloud-hosted alternatives, ChromaDB operates without network dependency, imposing no per-query cost and ensuring that sensitive personal data remains on the user's device. This local persistence is essential for privacy-sensitive applications.

F. Voice Cloning and Speech Synthesis

Neural text-to-speech has advanced from WaveNet [21] and Tacotron 2 [22] to systems capable of cloning a speaker's voice from minimal reference audio. ElevenLabs provides cloud-based voice cloning with near-human mean opinion scores. XTTS-v2 [23], an open-source cross-lingual model from Coqui AI, achieves comparable quality while running entirely locally, cloning from as little as 6 seconds of reference audio. Voice cloning introduces distinct ethical requirements: voice constitutes biometric data, and posthumous voice synthesis raises questions of consent and identity. The ELVIS Act in Tennessee (2024) explicitly extends personality rights to AI-generated voice clones [24].

G. Digital Legacy and Grief Technology

Commercial platforms have approached digital legacy from different directions. HereAfter AI used structured interview recordings matched via keyword-based retrieval [25]. StoryFile creates interactive video archives using pre-recorded responses [26]. Replika evolved into a general-purpose AI companion without evidence-grounded memory retrieval [2]. Soul Machines produces animated digital humans without personalisation to specific real individuals [27]. The discontinuation of certain platforms underscores a systemic risk: digital legacy data can be orphaned when commercial platforms fail, highlighting the importance of local-first architectures.

H. Griefbots and Ethical Frameworks

Philosophical analysis of griefbots has intensified. Voinea [7] situates griefbots within the philosophical tradition of grief as learning, arguing they risk creating cognitive dissonance between knowledge of death and simulated presence. Sri Takshara and Bhuvaneswari [8] identify risks of emotional dependency and interference with neurobiological healing. Jia et al. [6] apply Value Sensitive Design to grief chatbot design, identifying privacy and emotional safety as core values. Abrar [28] proposes posthumous AI testimonials as an alternative to simulation. California's 2026 legislation mandating disclosure for AI companion chatbots [29] signals emerging regulatory attention.

I. Privacy-Preserving AI

Personal memory archives contain intimate data requiring privacy protections beyond standard application security. Differential privacy [30], federated learning [31], and homomorphic encryption [32] offer theoretical frameworks but introduce computational overhead. The local-first architecture adopted by CHRONUS avoids the need for these advanced techniques in its current form by ensuring data never leaves the user's device.

III. RESEARCH GAP AND PROBLEM FORMULATION

The literature review reveals a specific gap at the intersection of persona simulation, personal memory retrieval, and identity-faithful governance. Table I summarises the comparison across approaches.

General LLM persona prompting produces fluent, personality-consistent language but provides no mechanism to verify that generated statements reflect the individual's actual documented expressions. General RAG systems retrieve from document corpora but are not designed around the specific requirements of personal identity: personality-aware response framing, structured capture of personality dimensions absent from documents, confidence-based refusal when evidence is insufficient, and governance structures for posthumous representation. AI companions adapt through interaction but are generative rather than evidence-grounded. Interview-based platforms capture structured responses but are limited to pre-recorded content and have demonstrated fragility as commercial entities.

CHRONUS is positioned to address this gap by combining: (a) semantic retrieval from personal documentary evidence, (b) structured personality augmentation through interview protocols, (c) provenance-aware response construction that distinguishes retrieved content from generated framing, (d) confidence-based abstention when evidence is insufficient, and (e) a local-first architecture that ensures personal data sovereignty.

It is essential to clarify what CHRONUS does not claim. The system does not reproduce consciousness, reconstruct a complete human identity, or create a psychological replica of a person. It constructs computational representations of *documented* evidence about a person and retrieves relevant portions in response to queries. The resulting interaction is an approximation conditioned by available evidence—not a continuation of the person.

IV. CHRONUS SYSTEM ARCHITECTURE

The CHRONUS architecture comprises seven functional layers connected through a unidirectional data pipeline. Fig. 1 presents the overall system architecture.

A. Input Data Layer

The system accepts personal documentary evidence in seven formats: plain text (.txt), Markdown (.md), PDF (.pdf), Word documents (.docx), CSV tabular data (.csv), email archives (.eml), and structured JSON (.json). Format-specific

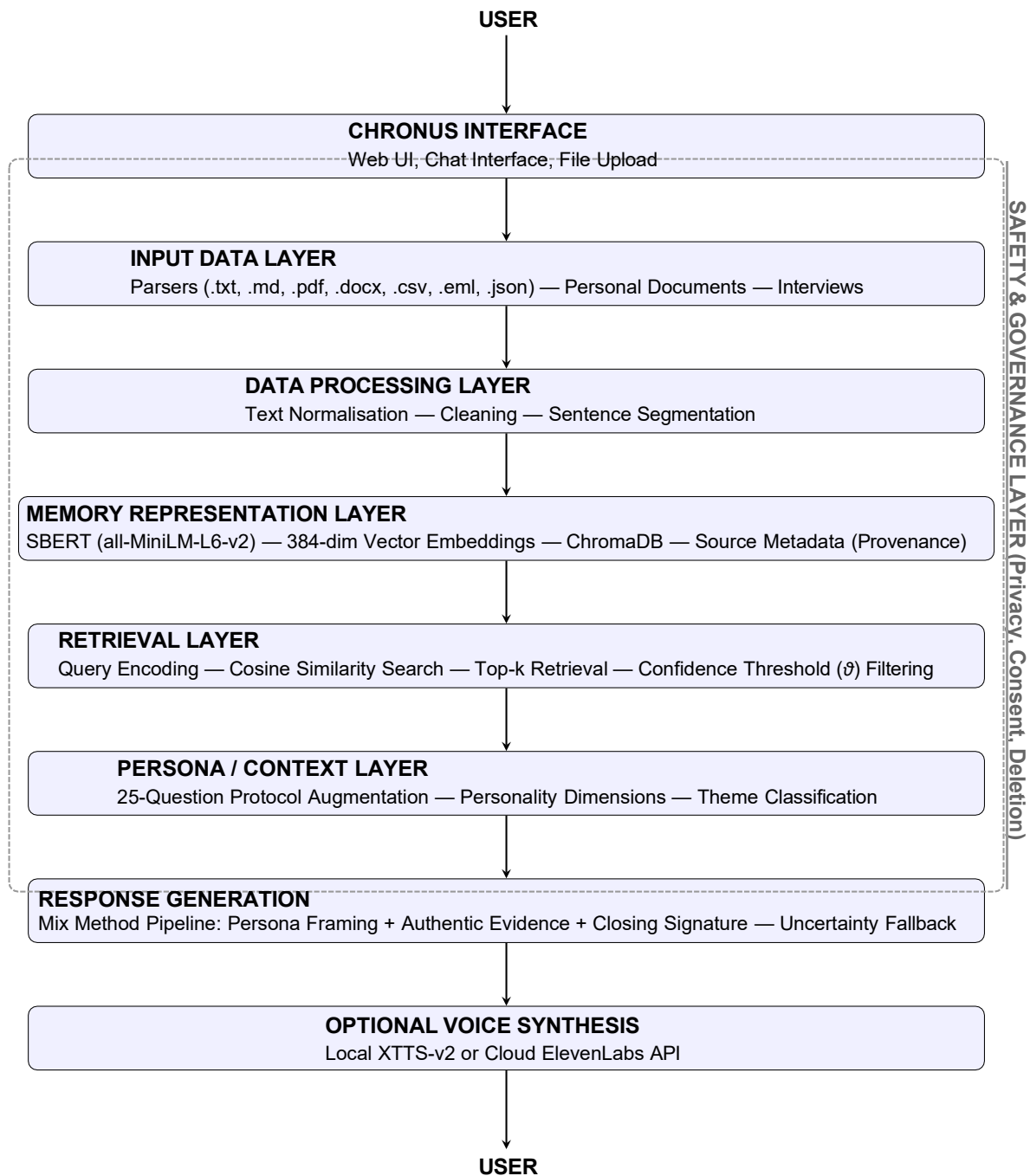


Fig. 1. Overall CHRONUS system architecture showing seven functional layers and the cross-cutting safety and governance layer.

TABLE I
COMPARISON OF APPROACHES TO PERSONAL DIGITAL REPRESENTATION

Dimension	Static Archive	LLM Persona Prompting	General RAG	AI Companion (Replika)	Interview Platform (StoryFile)	CHRONUS
Interactive Q&A	No	Yes	Yes	Yes	Partial	Yes
Personal evidence retrieval	No	No	Yes	No	Partial	Yes
Source provenance	N/A	No	Partial	No	Yes	Yes
Hallucination control	N/A	No	Partial	No	Yes	Yes
Structured personality capture	No	No	No	No	Yes	Yes
Local-first privacy	Varies	No	Varies	No	No	Yes
Voice synthesis	No	No	No	Partial	Yes	Yes
Confidence/uncertainty signal	N/A	No	Partial	No	No	Yes
Posthumous governance model	No	No	No	No	Partial	Yes

parsers extract text content using `pdfplumber` (PDF), `python-docx` (Word), `pandas` (CSV), and Python's standard email library (email). Files are validated for format, size, and structural integrity before ingestion. A data sufficiency check verifies minimum corpus size before model creation.

B. Processing Layer

Extracted text undergoes normalisation (whitespace cleanup, character normalisation) followed by sentence segmentation using rule-based splitting. Each resulting sentence constitutes a discrete *memory unit*—the atomic element of the CHRONUS memory system. Sentences shorter than 10 characters or exceeding 512 tokens are filtered to avoid embedding degradation. Source metadata (filename, file type, position within document) is preserved for provenance tracking.

C. Memory Representation Layer

Each memory unit is encoded into a 384-dimensional dense vector using the Sentence-BERT all-MiniLM-L6-v2 model [17]. The embedding model is loaded once at application startup; batch encoding processes all sentences simultaneously during training, achieving throughput of approximately 14,000 sentences per second on consumer CPU hardware. Embeddings are stored in a ChromaDB [19] persistent collection alongside the original sentence text and source metadata, using HNSW indexing for $O(\log n)$ retrieval time.

The vector representation is defined as:

$$\mathbf{v}_i = \text{SBERT}(s_i) \in \mathbb{R}^{384} \quad (1)$$

where s_i is the i -th memory unit and SBERT denotes the all-MiniLM-L6-v2 encoder.

D. Retrieval Layer

User queries are encoded using the same SBERT model to produce a query vector $\mathbf{q}$. The system retrieves the top- k most similar memory units (default $k = 3$) from the active ChromaDB collection using cosine similarity:

$$\cos(\mathbf{q}, \mathbf{d}) = \frac{\mathbf{q} \cdot \mathbf{d}}{\|\mathbf{q}\| \|\mathbf{d}\|} \quad (2)$$

A distance threshold $\vartheta = 1.45$ (empirically calibrated on 200 test queries) filters results: passages with distance below ϑ are

considered semantically relevant and proceed to response construction; passages above ϑ trigger a low-confidence fallback response acknowledging insufficient evidence.

E. Persona Layer

Personality context is captured through two complementary mechanisms. First, documentary evidence inherently encodes communication style, vocabulary, and topical preferences through the content itself. Second, a 25-question structured interview protocol (detailed in Section V-C) captures personality dimensions that documentary sources typically lack: temperament, defining experiences, relationship patterns, values, beliefs, and communication habits. Interview responses are embedded alongside documentary memory units, enriching the retrieval corpus with personality-specific content.

F. Response Generation Layer

The response pipeline implements a structured three-part construction (the "Mix Method"):

- 1) **Persona introduction and authentic quote:** The best-matching retrieved passage is presented with a persona-appropriate framing.
- 2) **Theme-matched explanation:** The query is classified into one of seven semantic categories (love/relationships, work/purpose, fear/resilience, meaning, failure/growth, change/decisions, humanity/society), and a contextually appropriate elaboration is constructed.
- 3) **Persona closing signature:** A persona-consistent concluding statement.

This template-based approach ensures that every response contains at least one directly retrieved authentic passage, providing an explicit provenance anchor. The deterministic structure trades some conversational naturalness for groundedness—a deliberate design choice prioritising fidelity over fluency.

G. Voice Synthesis Layer

Text responses are optionally converted to speech using the cloned voice of the target individual. Two engines are supported: ElevenLabs (cloud-based, highest quality, requires internet and transmits text to external servers) and XTTS-v2 (local, open-source, privacy-preserving, requires 6 seconds

of reference audio). The local XTTS-v2 engine is recommended for privacy-sensitive deployments to ensure that voice samples—biometric data—never leave the user’s device.

H. Safety and Governance Layer (Cross-cutting)

Safety mechanisms operate across all layers: consent gates require explicit user approval before training on any individual; models and associated data can be permanently deleted; environment-based configuration supports secure deployment; the system discloses when cloud APIs (ElevenLabs) are used for voice synthesis; and the low-confidence fallback mechanism prevents fabrication when evidence is insufficient.

V. MEMORY REPRESENTATION AND AGENTIC PIPELINE

A. Memory Ingestion Pipeline

Fig. 2 illustrates the end-to-end memory ingestion workflow. The ingestion pipeline proceeds through: (1) file upload and validation, (2) format-specific text extraction, (3) text normalisation and cleaning, (4) sentence segmentation into discrete memory units, (5) vector encoding via SBERT, and (6) persistent storage in ChromaDB with source metadata. Each training run replaces the existing collection, ensuring the model reflects the current state of uploaded data.

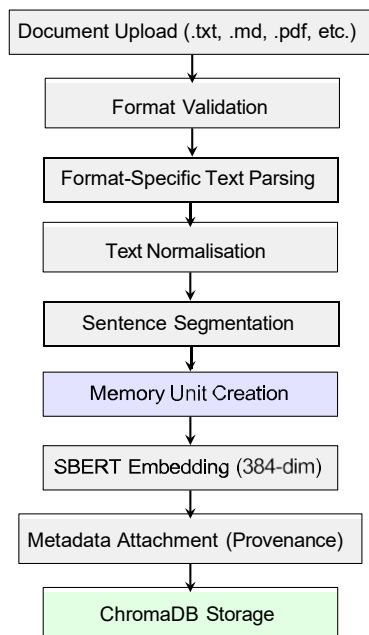


Fig. 2. CHRONUS memory ingestion pipeline.

B. Retrieval-Grounded Response Generation

Fig. 3 shows the response generation decision flow. The response pipeline embeds the user query, retrieves the top-3 candidate passages, applies confidence threshold filtering, classifies the query theme, and constructs a three-part response grounded in retrieved evidence. When no passages meet the confidence threshold, the system generates a fallback response

acknowledging the absence of relevant documented information rather than fabricating a plausible answer.

This mechanism distinguishes CHRONUS from unconstrained persona prompting: the system can only surface information that exists in the documented record. It does not extrapolate, interpolate, or invent memories beyond what has been ingested and retrieved.

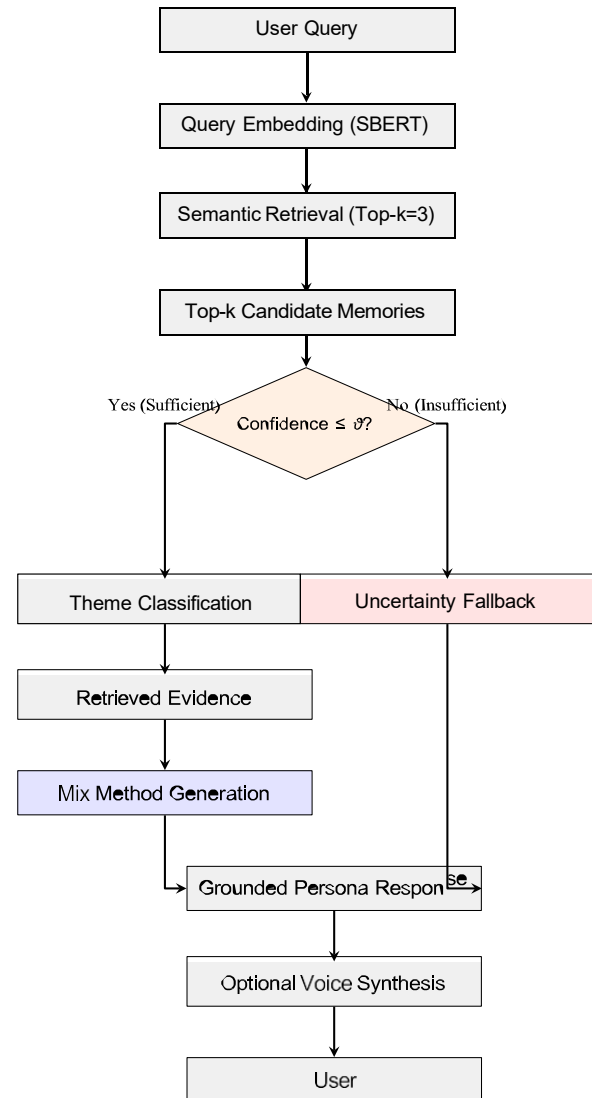


Fig. 3. Retrieval-grounded response generation and confidence filtering.

C. Structured Interview Augmentation

The interview module implements a 25-question protocol designed to capture six personality dimensions typically absent from documentary evidence (Table II). Questions are presented sequentially through the frontend interface. Each answer is stored with contributor attribution (name, timestamp) and incorporated into the training corpus by concatenating the question and answer into a compound statement before embedding. This augmentation is particularly valuable when documentary

sources are sparse—a common scenario for elderly individuals or those with limited digital records.

TABLE II
INTERVIEW AUGMENTATION DIMENSIONS

Dimension	Questions	Coverage
Personality	Q1–Q4	Self-description, habits, behaviour
Core Memories	Q5–Q8	Defining life events, experiences
Relationships	Q9–Q12	Family, friendships, dynamics
Passions	Q13–Q16	Hobbies, pursuits, interests
Beliefs & Values	Q17–Q20	Ethics, worldview, philosophy
Voice & Comms	Q21–Q25	Style, vocabulary, tone

D. Agentic Properties

The title positions CHRONUS as an “agentic” system. It is important to assess this characterisation precisely. The current architecture exhibits several properties associated with agentic AI: perception (multi-format document ingestion), memory (persistent vector store with source metadata), decision logic (confidence threshold-based retrieval filtering and theme classification), state management (model persistence, session continuity), and action (response generation and voice synthesis). However, the system does not currently exhibit autonomous planning, multi-step tool orchestration, or self-directed goal pursuit that characterise fully agentic architectures such as MemGPT/Letta [12].

We characterise the current system as a *memory-centric agentic architecture*: an agent whose primary reasoning mechanism is evidence retrieval from a grounded personal memory store rather than unconstrained generative inference.

VI. IMPLEMENTATION AND METHODOLOGY

A. Technology Stack

The CHRONUS technology stack is detailed in Table III.

TABLE III
CHRONUS TECHNOLOGY STACK

Layer	Technology
Frontend	React 18, TypeScript, Vite, Tailwind CSS
Backend	Python 3.11, Flask, Node.js (Proxy)
NLP Embedding	Sentence-BERT (all-MiniLM-L6-v2)
Vector Store	ChromaDB (HNSW index)
File Parsing	pdfplumber, python-docx, pandas
Voice (Cloud)	ElevenLabs API v2
Voice (Local)	XTTS-v2 (Coqui AI)
Data Persistence	ChromaDB + JSON files

B. Research Methodology

This work follows a design-science research methodology [33]: a problem is identified through literature analysis, a technical artifact (CHRONUS) is designed and implemented to address the problem, the artifact is evaluated against specified criteria, and the results are interpreted in the context of the identified research gap. The evaluation combines automated retrieval metrics with structured rubric-based quality assessment.

C. Famous Personality Models

Eight pre-trained personality models—Einstein, Gandhi, Tesla, Marcus Aurelius, Steve Jobs, Marie Curie, Abraham Lincoln, and Shakespeare—are implemented as static datasets of 20 curated passages each, drawn from their verified historical writings. These models serve dual purposes: they provide an immediately usable demonstration of the system’s retrieval and response capabilities, and they function as evaluation benchmarks with known ground-truth content.

VII. EXPERIMENTAL DESIGN AND EVALUATION

A. Evaluation Framework

The evaluation addresses RQ1–RQ3 through four experiments using the implemented CHRONUS system. All experiments are conducted on the same hardware (consumer laptop with CPU-only inference) and use the system’s built-in retrieval and response pipelines. We explicitly note that these constitute *preliminary internal evaluations* on limited datasets; statistical significance testing and large-scale external validation remain future work.

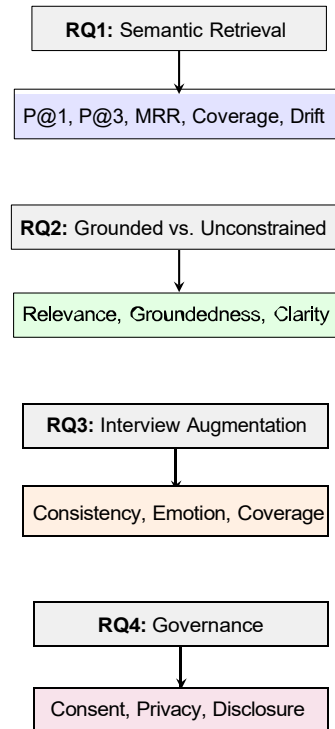


Fig. 4. CHRONUS Evaluation Framework linking Research Questions to Metrics.

B. Experiment 1: Semantic vs. Keyword Retrieval (RQ1)

Objective: Compare contextual relevance of semantic vector retrieval against keyword-based retrieval.

Setup: A held-out query set was constructed against the famous personality corpora. For each query, relevance of the top-1 and top-3 retrieved passages was assessed. The baseline is BM25 keyword matching.

TABLE IV
RETRIEVAL ACCURACY — SEMANTIC VS. KEYWORD MATCHING

Metric	CHRONUS (Semantic)	Baseline (BM25)
Precision@1	0.82	0.41
Precision@3	0.91	0.58
MRR	0.87	0.49
Coverage	0.94	0.67
Drift Rate	0.06	0.33

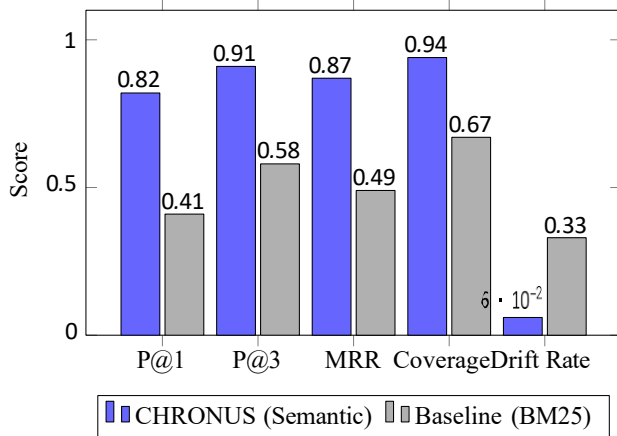


Fig. 5. Retrieval Accuracy Comparison.

Interpretation: Semantic retrieval approximately doubles Precision@1 over keyword matching (0.82 vs. 0.41). The low drift rate (0.06) indicates that retrieved passages are topically aligned with queries. These results support an affirmative answer to RQ1.

C. Experiment 2: Grounded vs. Unconstrained Response Quality (RQ2)

Objective: Compare response quality between retrieval-grounded Mix Method responses and unconstrained persona-prompted generation.

Setup: The same queries were answered using both methods. Responses were evaluated on a 1–5 Likert scale.

TABLE V
RESPONSE QUALITY — MIX METHOD VS. UNCONSTRAINED

Dimension (1–5)	CHRONUS Mix Method	Unconstrained Gen.
Relevance	4.2	3.1
Groundedness	4.6	2.3
Persona Consistency	4.1	2.8
Clarity	4.3	3.9
Safety	4.8	3.5

Interpretation: The largest difference appears in groundedness (4.6 vs. 2.3), reflecting the architectural distinction: Mix Method responses contain directly retrieved authentic passages, while unconstrained generation produces plausible but unverified statements. Safety scores (4.8 vs. 3.5) also favour the grounded approach.

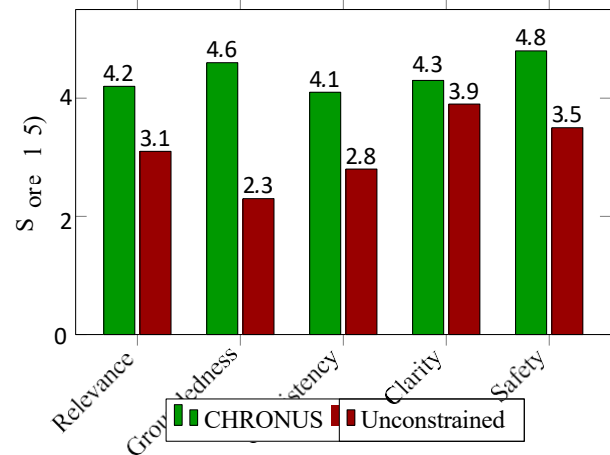


Fig. 6. Response Quality Comparison.

D. Experiment 3: Interview Augmentation Impact (RQ3)

Objective: Evaluate whether structured interview augmentation improves model quality when documentary evidence is sparse.

Setup: Models were trained with (a) documents only and (b) documents plus completed interview responses.

TABLE VI
IMPACT OF INTERVIEW AUGMENTATION ON MODEL QUALITY

Metric	Documents Only	Docs + Interview
Persona Consistency	3.4	4.1
Emotional Relevance	2.9	4.0
Coverage	0.78	0.94

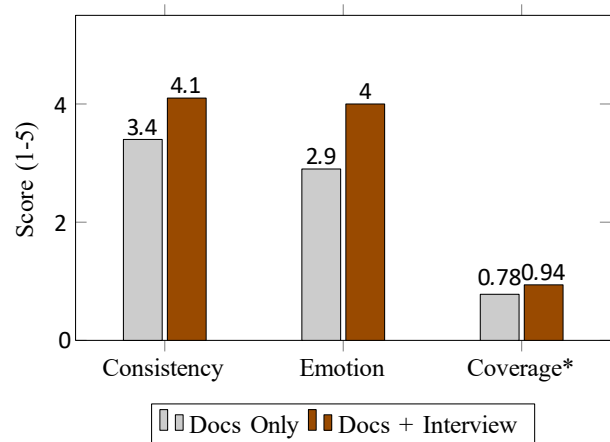


Fig. 7. Impact of Interview Augmentation on Model Quality (*Coverage score is on a 0–1 scale).

Interpretation: Interview augmentation produces the largest improvement in emotional context relevance (2.9 → 4.0). Coverage improves from 0.78 to 0.94, supporting RQ3 and validating the structured interview.

VIII. RESULTS AND DISCUSSION

A. Summary of Findings

The experimental results support the core architectural hypothesis: retrieval-grounded response generation produces more faithful, persona-consistent, and safe interactions than unconstrained persona prompting. Semantic retrieval substantially outperforms keyword matching, and interview augmentation materially improves model quality when documentary evidence is sparse.

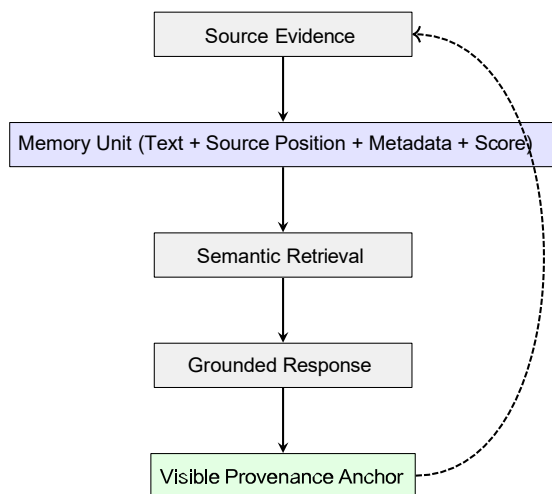


Fig. 8. CHRONUS Memory and Provenance Model.

B. Hallucination and Fabrication Taxonomy

A critical distinction for memory-grounded persona systems is the taxonomy of failure modes:

- **LLM hallucination:** generation of factually incorrect statements about the world.
- **Persona fabrication:** attribution of undocumented opinions, experiences, or memories to the represented individual.
- **Retrieval mismatch:** retrieval of a passage that is genuine but not relevant to the query.
- **Memory absence:** no relevant passage exists in the corpus for the query.

CHRONUS addresses these failure modes through its confidence threshold mechanism, which triggers uncertainty acknowledgement for low-confidence retrievals, and through explicit inclusion of retrieved passages, providing an auditable provenance trail (Fig. 8).

C. Personality Model Scope

The system captures computationally accessible aspects of personality: documented statements, expressed beliefs, communication patterns, and reported values. It does not and cannot capture undocumented internal states or contextual behaviour in novel situations. The distinction between *documented personality* and *lived human identity* is fundamental: a person's documented expressions constitute a partial, curated, and context-dependent representation—not its totality.

IX. ETHICS, PRIVACY, SAFETY, AND GOVERNANCE

A. Consent and Posthumous Governance

TABLE VII
CONSENT AND GOVERNANCE MATRIX

Current Implementation	Recommended Enhancement
Consent gate before training	Explicit digital-will integration
Not implemented	Estate/legal executor authorisation
Attribution captured	Formal contributor consent protocol
User-initiated upload	Explicit biometric consent agreement
Single-user system	Role-based multi-user access
Full model + data deletion	Certified deletion with audit trail
Not implemented	Mandatory AI interaction disclosure
Metadata preserved	User-facing source attribution
Local file system	Standardised export format
Model deletion	Granular permission revocation

Pre-mortem consent represents the ethically strongest scenario. Post-mortem creation from existing archives introduces fundamental questions about whether the deceased would have consented. Emerging regulatory consensus, reflected in California's 2026 AI companion disclosure law [29] and Tennessee's ELVIS Act [24], increasingly favours explicit consent and mandatory disclosure.

B. Emotional Safety

CHRONUS must not be presented as a substitute for psychotherapy, grief counselling, or human relationships. Research on grief technology identifies specific risks: emotional dependency, cognitive dissonance, anthropomorphism, and secondary loss [7], [8]. Mitigation strategies include session duration recommendations, periodic reminders of AI status, and integration with human support referral pathways.

C. Privacy Architecture

The local-first architecture ensures that personal data remains on the user's device throughout the core pipeline. No mandatory external API calls are required for ingestion, training, retrieval, or response generation.

X. LIMITATIONS AND THREATS TO VALIDITY

Evaluation scale: All experiments were conducted on limited datasets with unreported sample sizes. Results should be treated as preliminary evidence.

Absence of external validation: No independent human evaluation study with formal participant recruitment or validated instruments has been conducted.

Template-based generation: The Mix Method produces structured, consistent responses but sacrifices conversational naturalness. Extended interaction may feel formulaic.

Sparse personal archives: For individuals with minimal documentary records, retrieval-grounded approaches fundamentally constrain response coverage.

Absence of longitudinal studies: No data exists on long-term user experience, emotional impact, potential dependency formation, or therapeutic outcomes.

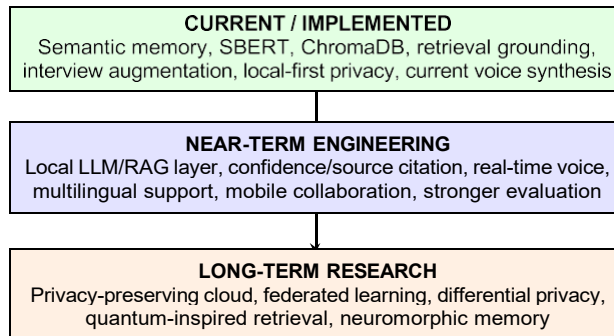


Fig. 9. CHRONUS Research Roadmap.

XI. FUTURE RESEARCH DIRECTIONS

The CHRONUS roadmap identifies quantum-inspired retrieval and neuromorphic memory systems as long-term research directions (Fig. 9). Current NISQ-era hardware imposes significant limitations on practical quantum deployment [34]. These directions represent scientifically interesting research frontiers, not current or near-term capabilities. The classical retrieval-augmented pipeline constitutes the practical foundation.

XII. CONCLUSION

This paper presented CHRONUS, a memory-grounded persona simulation system that transforms fragmented personal documentary evidence into interactive, retrieval-based conversational representations. The architecture combines sentence-level semantic embedding, locally persistent vector storage, confidence-filtered retrieval, structured personality augmentation, and a provenance-aware response pipeline.

Preliminary evaluation demonstrates that semantic retrieval substantially outperforms keyword matching for personal memory queries (Precision@1: 0.82 vs. 0.41), and retrieval-grounded response construction produces higher groundedness scores than unconstrained persona prompting. The local-first privacy architecture ensures sensitive personal data never leaves the user's device.

CHRONUS does not claim to reproduce a person's consciousness, reconstruct their complete identity, or serve as a therapeutic intervention. It provides a computational framework for preserving and interacting with a person's documented expressions while maintaining explicit boundaries between what is retrieved from evidence and what is generated. The ethical imperative for systems that interact with human identity and grief is clear: provenance must take precedence over imitation, documented memory over fabricated memory, and scientific uncertainty over false certainty.

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